



Autoignition of CRC diesel surrogates at low temperature combustion conditions: Rapid compression machine experiments and modeling

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ABSTRACT

As federal programs require increasingly stringent engine emissions and fuel economy standards, these ambitions can only be met if next-generation combustion technology is developed focusing on high-efficiency and low-emissions engines. Recent research has indicated the need to operate engines at higher compression ratios and with low temperature combustion (LTC) to achieve the needed gains in engine efficiency and reductions in emissions. Because there is a lack of understanding of the chemistry of diesel fuel components and their mixtures at these LTC conditions, this limits the ability to develop predictive chemical kinetic models that can be used to optimize engine combustion. The current study aims to fill in gaps in fundamental combustion data on surrogate fuel mixtures relevant to diesel fuels. Specifically, four multicomponent diesel surrogates formulated by the Coordinating Research Council (CRC) to emulate an ultra-low-sulfur research-grade #2 certification diesel fuel (CFA), namely V0a (4 components), V0b (5 components), V1 (8 components), and V2 (9 components), have been investigated in a rapid compression machine (RCM) through determination of total and first-stage ignition delay times. Autoignition characteristics of lean to rich fuel/O₂/N₂ mixtures, for the four CRC surrogates and CFA, have been measured using an RCM at LTC relevant pressures and temperatures, in the ranges of 10–20 bar and 650–1000 K, respectively. The equivalence ratios have been varied by independently changing the oxygen mole fraction and the fuel mole fraction in the test mixtures, thereby illustrating the individual effects of oxygen concentration and fuel loading on diesel autoignition. Autoignition results of these four CRC surrogates are compared among them and with those of CFA. Some degree of agreement in autoignition response between each CRC surrogate and CFA is observed, while discrepancies are also identified and discussed. In addition, a detailed chemical kinetic model for diesel surrogates has been developed and validated against these newly-acquired RCM data. This model shows reasonable agreement with the overall ignition delay time results of the current RCM experiments. Chemical kinetic analyses of the developed model were further conducted to help identify the reactions controlling the autoignition processes and the consumption of fuel components in CRC surrogates.

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1. Introduction

Because of increasingly stringent engine emissions and fuel economy standards, there is an urgent need for developing future diesel engines with higher efficiency and lower emissions. Therefore, low temperature combustion (LTC) is currently being pursued to develop new types of advanced engines such as homogeneous charge compression ignition (HCCI) and premixed charge compression ignition (PCCI) engines. These LTC engines are de-

signed to operate at lower temperatures, greater degree of premixing, and higher dilution levels than conventional diesel engines to reduce the emission of nitrogen oxides (NO_x) and particulate matter (PM), and to burn the fuel more efficiently [1]. Unlike conventional diesel combustion, the overall reactivity leading to ignition is constrained, which results in increasing dominance of fuel chemistry in such LTC modes. As a result, the understanding of the autoignition characteristics of diesel fuels under low-to-intermediate temperatures becomes important. In order to achieve the goal of higher efficiency and lower emissions diesel engines, both experimental and computational investigations need to be conducted. Experimentally, the autoignition results of diesel fuels at low-to-intermediate temperature will help in developing a comprehensive

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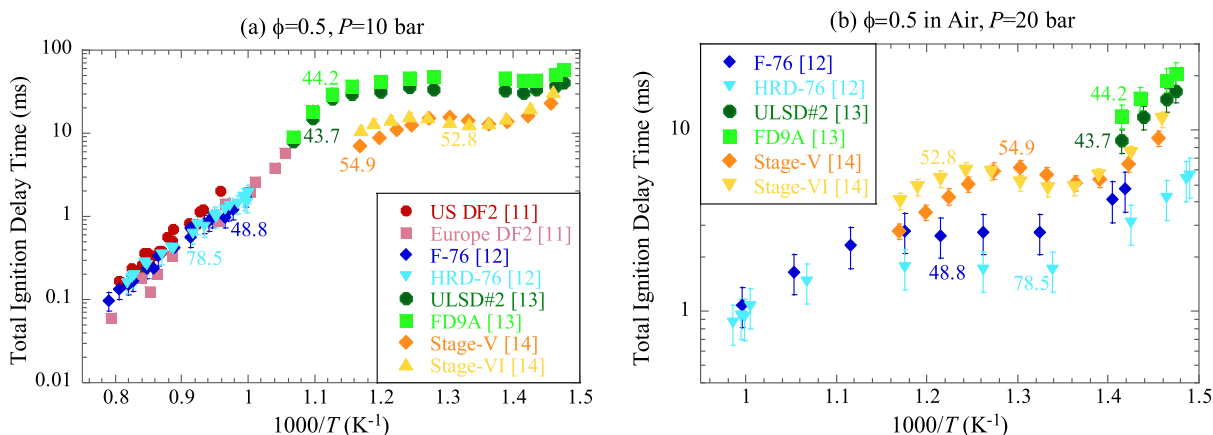


Fig. 1. Comparison of experimental total ignition delay times of diesel fuels from literature studies [11–14] with colored numbers representing the reported DCN values of diesel fuels. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

understanding of diesel ignition, as well as providing a validation database for model development. Computationally, a comprehensive chemical kinetic model of diesel is imperative for accurate prediction of ignition and emissions characteristics of diesel engines. Because diesel fuels contain hundreds, even thousands of species, the composition of diesel is too complex to model. Therefore, it is necessary to develop a surrogate fuel, which is a simpler mixture that can capture the essential chemical/physical properties and performance characteristics of the target diesel fuel to sufficient accuracy, but contains approximately 10 or fewer pure “palette” compounds [2] for the sake of computational tractability. As such, the development of diesel surrogates and their chemical kinetic models is vital in reducing the design cycle of future diesel engines.

Since diesel fuel has limitations when operating in HCCI and PCCI engines because of its relatively high reactivity compared to gasoline, ignition phasing control can be difficult as ignition timing is related to the chemical kinetics of the fuel/air mixture. As a result, several strategies have been proposed in recent studies such as running gasoline in diesel engines [3] and employing diesel/gasoline blends in advanced combustion modes [4–6]. Evidently, ignition phasing control can be greatly benefited by having good understanding of both diesel and gasoline autoignition at LTC conditions. As the autoignition of gasoline and its surrogates has been extensively studied with the most up-to-date detailed chemical kinetic model development [e.g., 7] and further experimental ignition delay time data validation [e.g., 8], similar progress in diesel experiments and surrogate modeling is required. Hence, the emphasis of this study is on experimental autoignition results of diesel surrogates at low-to-intermediate temperatures and the development of a detailed diesel surrogate model.

There are a few studies focusing on autoignition characteristics of diesel fuels. Spadacinni and Tevelde [9] studied the ignition delay characteristics of No.2 diesel and a research grade diesel in a continuous flow reactor at pressures of 10–30 atm, temperatures up to 1000 K, and fuel/air equivalence ratios of $\phi=0.3$ –1.0. Penyazkov et al. [10] investigated the high temperature (1065–1838 K) oxidation of an U.S. commercial diesel (DF-2) in a pre-heated shock tube at $\phi=0.5$ –2.0 in air at pressures of 4.7–10.4 atm. Subsequently, Haylett et al. [11] studied the ignition of four diesel blends in a shock tube at equivalence ratios of 0.2–1.0, with pressures ranging from 1.7 to 8.6 atm, and in the temperature range of 947–1261 K by varying O_2/Ar ratios. Gowdagiri et al. [12] measured ignition delay times for petroleum (F-76) and algae-derived diesel blends (HRD-76) in a shock tube over temper-

atures of 671–1266 K, pressures of 10 atm and 20 atm, and equivalence ratios of 0.5 and 1.0. More recently, Kukkadapu and Sung [13] investigated the ULSD#2 and FD9A diesel fuels using a heated rapid compression machine (RCM), with the ignition delay time measurements being conducted at compressed pressures of 10 bar, 15 bar, and 20 bar covering low-to-intermediate temperatures of 678–938 K, with diesel/ O_2/N_2 mixtures at varying equivalence ratios of $\phi=0.5$, 0.69, and 1.02. Yu et al. [14] then studied the autoignition characteristics of two commercial diesels (China Stage-V and Stage-VI) in a heated RCM at equivalence ratios of 0.37–1.25, pressures of 10–20 bar, and temperatures of 687–865 K.

The experimental total ignition delay times of diesel fuels reported in [11–14] for $\phi=0.5$ (in Ar-air [11] and in air [12–14]) are summarized and compared in Fig. 1. It is noted that the shock tube data shown in Fig. 1 are scaled to 10 bar and 20 bar using the pressure dependence discussed in the respective studies [11,12]. In addition, the cetane ratings, such as cetane number (CN, determined by ASTM D613), derived cetane number (DCN, determined by ASTM D6890), and cetane index (CI, determined by ASTM D976), and compositions of these diesel fuels are listed and compared in Table 1. It can be concluded that the total ignition delay times overlap well for all diesel fuels investigated at temperatures higher than 1000 K. At lower temperatures, the total ignition delay time differences can be attributed to different contents of saturates and aromatics among the diesel fuels tested. The ranking of ignition reactivity for these diesel fuels is also consistent with their DCNs. Thus, these comparisons help understand the effects of cetane rating and fuel composition on diesel ignition propensity, and further assist the formulation of diesel surrogates as well as the development of surrogate models.

Besides those past works on autoignition of diesel blends that have provided data over a wide range of pressures and temperatures, there are also studies focusing on the ignition characteristics and formulation of diesel surrogates. For the works of diesel surrogate formulation, most studies proposed surrogate mixtures containing a small number of pure compounds and focused on matching the oxidation, flame extinction, and sooting levels of the target diesel fuel. *n*-Heptane has been widely used as a single-component diesel surrogate in multidimensional computational fluid dynamics modeling studies because of its cetane number being comparable to those of actual diesel fuels [15–20]. As actual diesel fuels consist of longer carbon chain structures, *n*-hexadecane has then become a promising component [21–26]. However, as single-component diesel surrogate models have limitations such as the Hydrogen/Carbon Molar Ratio (H/C) cannot match that of

Table 1

Comparison of cetane ratings and compositions of diesel fuels from literature studies [11–14].

Property/Composition	US DF2 [11]	Europe DF2 [11]	F-76 [12]	HRD-76 [12]	ULSD#2 [13]	FD9A [13]	Stage-V [14]	Stage-VI [14]
DCN	–	–	48.8	78.5	43.7	44.2	54.9	52.8
CN	–	–	–	–	43.3	43.9	52.5	52
CI	43	55	44	78	–	–	–	–
Saturates (weight%)	–	–	79	99	69.85	62.6	83.8	84.2
<i>n</i> -Alkanes	–	–	54	48	16.38	10.26	49.2	40.4
isoAlkanes	–	–	19	51	18.44	23.4	–	–
Cycloalkanes	–	–	6	~0	35.03	28.94	34.6	43.8
Aromatics (weight%)	–	–	16	~0	30.05	36.43	16.2	15.8

Table 2

Compositions of four CRC diesel surrogates [2].

Palette Compound	Abbreviation	V0a (mole%)	V0b (mole%)	V1 (mole%)	V2 (mole%)
<i>n</i> -hexadecane	NC16H34	27.8	0	2.7	0
<i>n</i> -octadecane	NC18H38	0	23.5	20.2	10.8
<i>n</i> -eicosane	NC20H42	0	0	0	0.8
iso-cetane (2,2,4,4,6,8,8-heptamethylnonane)	HMN	36.3	27.0	29.2	0
2-methylheptadecane	2MHPD	0	0	0	7.3
<i>n</i> -butylcyclohexane	NBCH	0	0	5.1	19.1
1,3,5-triisopropylcyclohexane	TIPCX	0	0	0	11.0
<i>trans</i> -decalin	DECALIN	14.8	0	5.5	0
perhydrophenanthrene	PHP	0	0	0	6.0
1,2,4-trimethylbenzene	T124MBZ	0	12.5	7.5	0
1,3,5-triisopropylbenzene	TIPB	0	0	0	14.7
tetralin	TETRA	0	20.9	15.4	16.4
1-methylnaphthalene	A2CH3	21.1	16.1	14.4	13.9

the target diesel fuel, and the formation of polycyclic aromatic hydrocarbons in diesel fuel combustion cannot be well described by a single-component diesel surrogate fuel model [27], it is necessary to develop surrogate models with the fuel compositions that better represent the actual diesel fuels. Therefore, aromatics are usually used as the second component in two-component diesel surrogates to account for soot formation. There are also multi-component diesel surrogates proposed in literature, with most of them containing 4–5 components, and with each component representing a hydrocarbon class in real diesel fuels. Pitz and Mueller [28] conducted a literature review on studies of diesel surrogate fuels before 2011, starting with binary fuel blends and then progressing to more complex surrogate mixtures. In addition, Reyes et al. [29] developed a method to determine the ignition delay times of diesel surrogate fuels using data from a constant volume combustion bomb, with *n*-heptane and a mixture of 50% *n*-heptane and 50% toluene by mass as the test fuels because both have been proposed as diesel fuel surrogates under the Homogeneous Charge Compression Ignition (HCCI) engine mode. Carr et al. [30] studied the combustion characteristics of binary mixtures of *n*-hexadecane and toluene in a single-cylinder research engine. Moreover, Pei et al. [31] developed a detailed chemical kinetic model by combining two previously developed reaction mechanisms of *n*-dodecane [32] and xylenes [33] as a mixture of *n*-dodecane and *m*-xylene was investigated as a diesel surrogate for compression ignition engine applications. For recent studies of multi-component diesel surrogates, Chang et al. [34] developed two diesel surrogates with the same fuel components, including *n*-decane, isooctane, methylcyclohexane, and toluene, to represent straight-alkane, branched-alkane, cycloalkane, and aromatic compounds in diesel fuels, respectively, but the compositions of the two diesel surrogates are different in order to represent different diesel fuels. Poon et al. [35] developed and validated reduced chemical kinetic models for several hydrocarbons, including *n*-hexadecane, 2,2,4,4,6,8,8-heptamethylnonane, cyclohexane, and toluene. Subsequently, Poon et al. [36] presented and discussed multi-component diesel surrogate models with blends of various fuel components. In general, it can be concluded from these stud-

Table 3

Comparison of DCN values of CFA and four CRC diesel surrogates [2].

DCN	CFA	V0a	V0b	V1	V2
measured ^a	43.7 44.9	46.2	44.4	45.5	42.6
predicted ^b	–	43.7	43.5	44.1	43.5

ies that the number of fuel components in a surrogate model was limited at first, but surrogate models with greater number of fuel components were then established due to the rapid advancement in chemical kinetics and computing power.

Mueller et al. [37] provided a methodology for formulating diesel surrogate fuels with improved fidelity in matching the molecular structures, ignition quality, and distillation characteristics of two target diesel fuels, a 2007 emissions certification diesel fuel and a Coordinating Research Council (CRC) Fuels for Advanced Combustion Engines (FACE) diesel fuel. In a follow-up study by the CRC, Mueller et al. [2] further formulated, blended, and characterized a set of four diesel surrogate fuels, including 4-component V0a, 5-component V0b, 8-component V1, and 9-component V2, that were designed to emulate the certified ultra-low-sulfur research-grade diesel, CFA, supplied by Chevron Phillips Chemical. Table 2 lists the compositions of the four CRC diesel surrogates taken from Mueller et al. [2], with the abbreviations of palette compounds based on the newly developed chemical kinetic model of diesel surrogates, which will be introduced in due course. For the fuel components not included in the current surrogate model, their abbreviations follow those used in [2]. In addition, the DCN values of CFA and four CRC diesel surrogates, taken from Mueller et al. [2], are listed in Table 3 to quantify and compare their ignition quality.

In this work, an RCM has been used to obtain the autoignition data of lean to rich fuel mixtures at engine relevant conditions for the four CRC diesel surrogates. The autoignition results of the CRC surrogates are also compared with those of CFA at the same test conditions to determine how well the surrogates

represent and emulate CFA based on ignition delay times and pressure traces. In addition, the effects of pressure, temperature, fuel loading, and oxygen concentration on autoignition of CFA and its surrogates have been investigated herein. Furthermore, we also aim to develop a detailed diesel surrogate chemical kinetic model including most of the key components listed in Table 2 and validate it by comparing the simulated results with the experimental data of the CRC diesel surrogates. While the current efforts are not intended to investigate and simulate multi-phase/heterogeneous diesel ignition/combustion in typical CI engines, gaining the fundamental understanding of the autoignition chemistry for diesel surrogates will be valuable in future studies of real diesel fuels at engine conditions.

2. Experimental specifications

A rapid compression machine is used to compress gas mixtures to elevated pressures and temperatures using a fast moving piston which is brought to rest towards the end of compression. Post compression, the piston is held stationary thus providing a constant volume reactor. This RCM is designed to reduce bulk fluid motion and thermal in-homogeneities, thereby providing an ideal platform for studying chemical kinetics and ignition characteristics of a fuel. The present RCM consists of a pneumatic cylinder, a hydraulic cylinder, and a reaction chamber, in which the reactant gas mixture is compressed by a creviced piston arrangement during each experimental run. First, the creviced piston in the reaction chamber is retracted from the top dead center to the bottom dead center. It is held in place by pressurized oil in the hydraulic cylinder. When the solenoid valve is triggered, the pressure in the hydraulic chamber is released and the creviced piston moves forward, driven by the high-pressure air tank and the pneumatic piston inside the pneumatic cylinder. The creviced piston is then decelerated and stopped by a stopping ring and groove mechanism. The compression stroke of the RCM typically takes a time in the range of 25–50 ms. A thermal-shock resistant, dynamic pressure transducer (Kistler 6125C) is equipped on the reaction chamber for dynamic pressure measurements during the experiment. The compression ratio of this RCM can be varied by changing the length of stroke or/and the clearance volume. Here, split shims are used to vary the clearance between hydraulic cylinder and the reaction chamber, while the stroke can be adjusted by using spacers. More details about the RCM design and operation can be found in [38,39].

Each of the diesel/oxidizer and surrogate/oxidizer mixtures is prepared in a stainless steel mixing tank. A magnetic stirrer is placed at the center of the tank bottom to provide a mixing mechanism. A rupture disc is used as a safety measure against accidental overpressure within the tank. For the current study, the mixture is prepared at room temperature, with CFA diesel fuel or surrogate fuel injected directly inside the mixing tank using a glass syringe, followed by oxygen and nitrogen filled into the tank on a barometric measure in the desired $O_2:N_2$ molar proportion. Recognizing that the gas heating effect can occur during the tank filling process as the line enthalpy is converted to tank internal energy, we always ensure filling each gas to the target pressure while the tank temperature is within $\pm 1.5^\circ C$ of the room temperature. The mixing tank, the manifold, and the reaction chamber, all equipped with heating tapes, are heated to the desired preheat temperature of 420 K for 4 h before starting the experiments to ensure complete vaporization of CFA diesel or surrogate fuel and a homogeneous mixture. For ensuring that the CFA diesel fuel injected to the mixing tank is completely vaporized, the amount of fuel injected has been calculated to warrant that the partial pressure of the fuel/oxidizer mixture is less than 14 Torr when preheating the mixing tank to 420 K, which has been indicated and

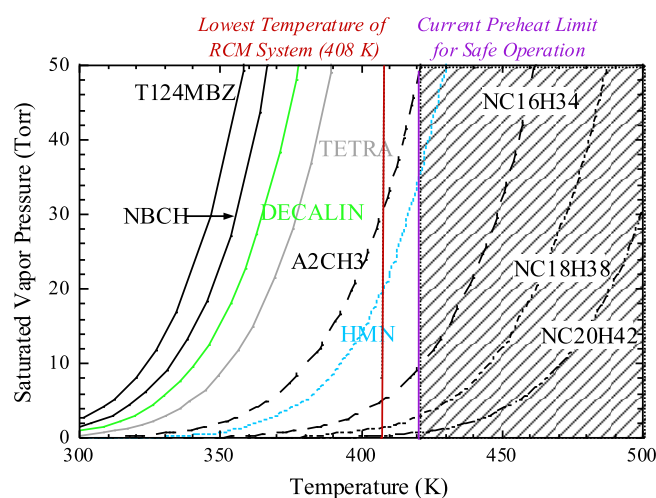


Fig. 2. Plot showing the saturated vapor pressures of CRC surrogate components as a function of temperature. See Table 2 for the abbreviations of the component names. (For interpretation of the references to color in this figure, the reader is referred to the web version of this article.)

discussed in [13]. For the four CRC surrogates, we ensure that the partial pressure of each surrogate component is less than half of its saturated vapor pressure at the preheat temperature of 420 K. In order to monitor the wall temperature uniformity of the reaction chamber, three thermocouples are used and evenly distributed along the outside wall of the reaction chamber. For the set preheat temperature of 420 K in the current study, the temperature readings of the three thermocouples are typically within 413–423 K. Furthermore, a pressure transducer is located at the manifold for mixture preparation. This pressure transducer is also heated, with a thermocouple placing on the transducer head to ensure that the pressures recorded are under the desired preheat temperature (420 ± 1 K), and hence condensation inside the transducer head can be avoided.

Figure 2 shows the saturated vapor pressures of the diesel surrogate components as a function of temperature. The compositions of the four CRC surrogates, as well as the abbreviations of the component names can be found in Table 2. All the components of V0a, V0b, and V1 are included in Fig. 2, while NC20H42 only exists in the V2 surrogate. The purple line in Fig. 2 represents the preheat temperature of 420 K, and the red line in Fig. 2 indicates the lowest temperature of the RCM system for diesel fuel experiments, which is conservatively set to be 408 K. When heating the RCM to 420 K, the temperature readings of all temperature-measuring points on the mixing tank, reaction chamber, and manifold are within 413–423 K. Therefore, 408 K is selected as the lowest temperature limit to ensure no fuel condensation. This also means that when preheating the mixture before experiments, a margin of 12 K is chosen to warrant that the partial pressure of each surrogate component is still lower than its saturated vapor pressure at 408 K. It can be observed from Fig. 2 that in this temperature range, when designing a test matrix, the top five components require attentions to avoid condensation are NC20H42, NC18H38, NC16H34, HMN, and TETRA, because of their relatively lower saturated vapor pressures.

The compressed temperature (T_c), namely the gas temperature at the end of compression, is used as the reference temperature in reporting the present RCM data, and is deduced from the measured pressure trace by using the assumption of “adiabatic core hypothesis” [40]. Under this hypothesis, it is assumed that the heat transfer from the core volume of the reaction chamber only occurs in a thin boundary layer near the wall, and the central core region

Table 4
Test conditions for CFA diesel and CRC surrogates.

ϕ	Oxidizer	X_{fuel} (%)	X_{O_2} (%)	X_{N_2} (%)	P_C (bar)
0.5	Air	0.51	20.9	78.59	10
0.69	Diluted	0.51	15.12	84.37	10
1.02	Diluted	0.51	10.24	89.25	10
1.02	Diluted	0.51	10.24	89.25	15
2.0	Diluted	0.51	5.12	94.37	20
0.7	Diluted	0.36	10.47	89.17	15

Table 5
Saturated vapor pressures of some components at 420 K and 408 K and their partial pressures calculated based on the pressure in the mixing tank of 1280 Torr at 420 K with the fuel molar percent of 0.51%.

Component Name Abbreviation	Saturated Vapor Pressure at Preheat 420 K (Torr)	Saturated Vapor Pressure at Preheat 408 K (Torr)	Maximum Partial Pressure (Torr)
A2CH3	49.16	31.42	1.39
HMN	33.28	20.02	2.39
NC16H34	9.03	5.06	1.83
NC18H38	3.1	1.66	1.55
NC20H42	1.12	0.57	0.05

is adiabatic by modeling the heat transfer as an effective reduction in the geometric compression ratio [41,42]. Thus, the compressed temperature can be calculated by $\ln\left(\frac{P_C}{P_0}\right) = \int_{T_0}^{T_C} \frac{\gamma}{T(\gamma-1)} dT$, where P_C is the pressure at the end of compression, P_0 is the initial pressure, T_0 is the initial temperature, and γ is the temperature-dependent specific heat ratio.

2.1. Test matrix design and limitations

The test conditions investigated in the current study are outlined in Table 4, which have been designed to demonstrate the effects of pressure, fuel loading, and oxygen concentration on total and first-stage ignition delay times, and to cover a wide range of temperature windows in the Arrhenius plot, including low temperature regime (LTR), negative temperature coefficient (NTC) regime, and intermediate temperature regime (ITR). The equivalence ratios (ϕ), molar percentages of mixture constituents (X_i), and compressed pressures (P_C) tested are specified in Table 4. The temperature windows covered in Arrhenius plots for experimental ignition delay times of each set of (ϕ , X_i , P_C) have been shown in Table S1 of Supplementary Material.

It should be mentioned that the equivalence ratios specified in this paper are calculated based on CFA diesel and can also be regarded as the equivalence ratios for CRC surrogates, because the differences in equivalence ratios among CFA diesel and four CRC surrogates as a result of the compositional differences are less than 0.1. The mixture conditions, varying from fuel lean, stoichiometric, to fuel rich while avoiding the complications of soot production in the experiments, are investigated herein in order to provide experimental data over a wider equivalence ratio range for chemical kinetic model development and validation.

It is also noted that not all temperature windows can be covered in Arrhenius plots for certain test conditions, which is because of the limitations such as partial pressure, ignition delay time, and exothermicity in the compression stroke. These limitations are explained in the following with V0b as an example, as it includes the highest amount of NC18H38 (23.5% mole percentage) compared with the other three CRC surrogates. Table S2 in SM further summarizes the experimental conditions of V0b and CFA, as well as the limitations for not being able to cover the entire curve of Arrhenius plot for V0b. As mentioned earlier, when preparing a fuel/oxidizer mixture, we always ensure that the partial pressure of each component in V0b is less than half of its corresponding saturated pressure at the preheat temperature of 420 K. Since the highest fuel molar percentage (X_{fuel}) is set to be 0.51%, when heating from room temperature to 420 K, the highest pressure of the test mixture in the mixing tank can be 1280 Torr, with the saturated pressure and partial pressure of *n*-octadecane being 3.1 Torr and 1.54 Torr, respectively. When $P_C=15$ bar, the required P_0 for data points in the NTC regime is typically in the range of 900–1300 Torr. Hence, only a part of NTC can be covered for $X_{\text{fuel}}=0.51\%$ at $P_C=15$ bar, with the lowest temperature data point starting at $T_C=780$ K with $P_0=1050$ Torr. The data points in the LTR are not available for $X_{\text{fuel}}=0.51\%$ at $P_C=15$ bar as the required P_0 exceeds 1300 Torr. When $P_C=20$ bar, the required P_0 for data points in the

LTR and NTC regime is typically in the range of 1300–2200 Torr. As a result, experimental data of V0b in the LTR and NTC regime are not available for test conditions with $X_{\text{fuel}}=0.51\%$ at $P_C=20$ bar.

Since the residual vortex (if any) could lead to inhomogeneity in the core region of the reaction chamber after long experimental duration [43], we choose not to report data points that have total ignition delay times much longer than 100 ms. For $\phi=1.02$ with $X_{\text{fuel}}=0.51\%$ at $P_C=10$ bar, the ignition delay data of V0b are not available because based on the autoignition results of V0b at $\phi=0.69$ and $P_C=10$ bar, as well as the CFA results at $\phi=1.02$ and $P_C=10$ bar, it can be inferred that data points of V0b in both the LTR and NTC regime will exceed 100 ms.

Furthermore, the NTC regime is not available for $\phi=0.35$ with $X_{\text{fuel}}=0.36\%$ at $P_C=15$ bar, because of some exothermicity in the compression stroke is noted from the pressure trace comparison of reactive experiment and its nonreactive counterpart (obtained by replacing O_2 with N_2 in the test mixture) with the same initial pressure and machine settings. Figure S1(a) in SM shows an example of pressure trace with exothermicity near the end of the compression stroke, marked by a red rectangle, by comparing the reactive and nonreactive experimental pressure traces with the same initial pressure and machine settings. It can be observed that there is a discrepancy near the end of compression, showing that the reactive experiment can reach higher pressure and temperature at the end of compression as compared to the nonreactive experiment. By examining the reactive and nonreactive pressure traces for each RCM experiment, those ignition delay data with exothermicity during the compression stroke are not included in the reported Arrhenius plots.

2.2. Vaporization checks of CFA, V0a, and V0b

The vaporization checks have been conducted to ensure that the liquid fuel injected into the mixing tank have vaporized completely. Theoretically, Table 5 shows the saturated vapor pressures and partial pressures of the five components mentioned earlier that have relatively lower saturated vapor pressures at preheat temperatures of 420 K and 408 K. This table is calculated based on the highest pressure in the mixing tank, namely 1280 Torr at 420 K with the highest fuel molar percentage of 0.51%. It can be observed from Table 5 that the partial pressure of each component is less than half of its corresponding saturated vapor pressure at 420 K, and is also less than its corresponding saturated vapor pressure at 408 K.

Experimentally, the vaporization checks were carried out by preheating the mixing tank to 420 K, then injecting a known mass of fuel and recording the time history of the pressure inside the tank. The pressure calculated from the ideal gas law for the same known fuel mass was set as a reference, which assumes that the fuel has completely vaporized in the mixing tank. Figure 3 shows the vaporization check results of CFA, V0a, and V0b, with their

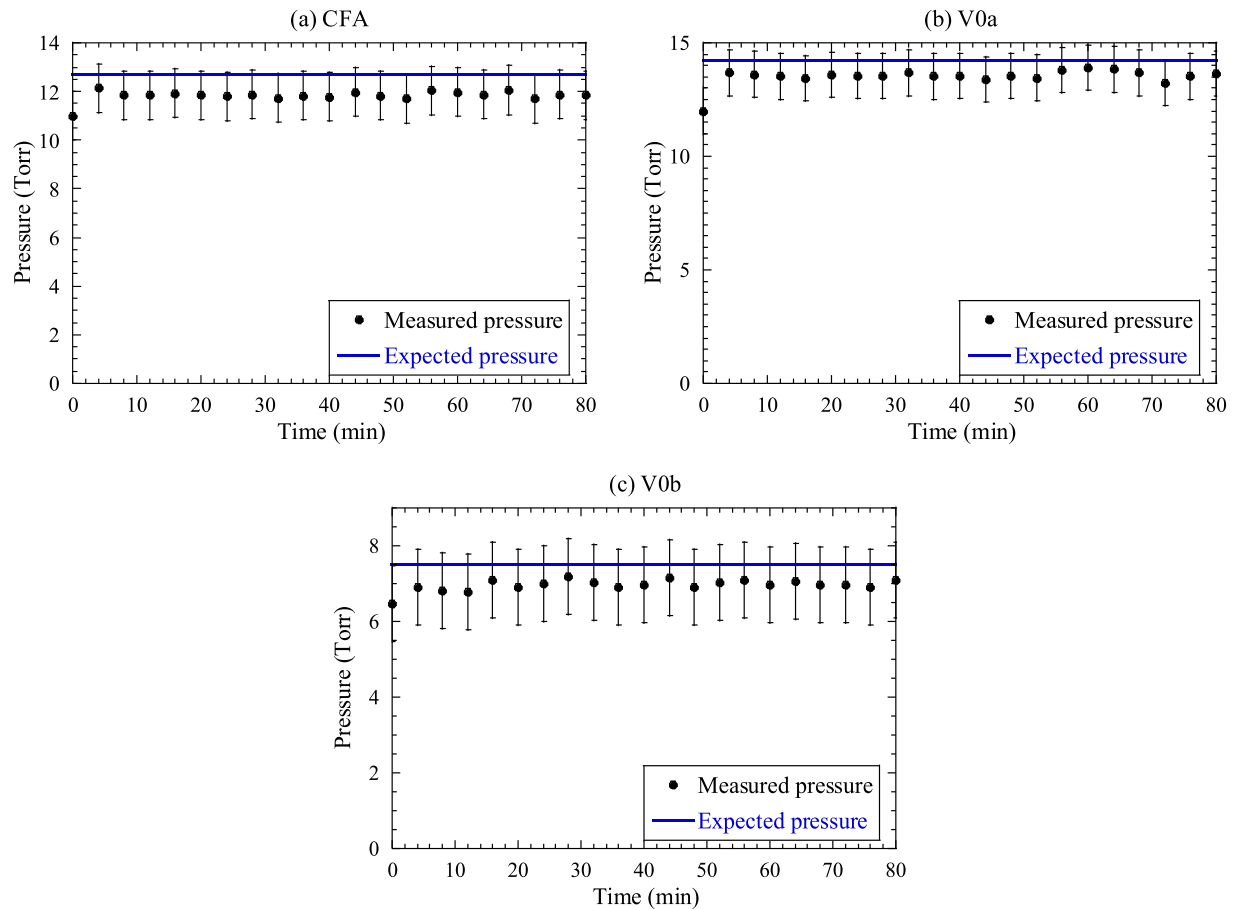


Fig. 3. Measured time-history of pressure in the mixing tank after liquid fuel injection for (a) CFA, (b) V0a, and (c) V0b. The pressure calculated from the ideal gas law assuming complete fuel vaporization is also shown as a reference.

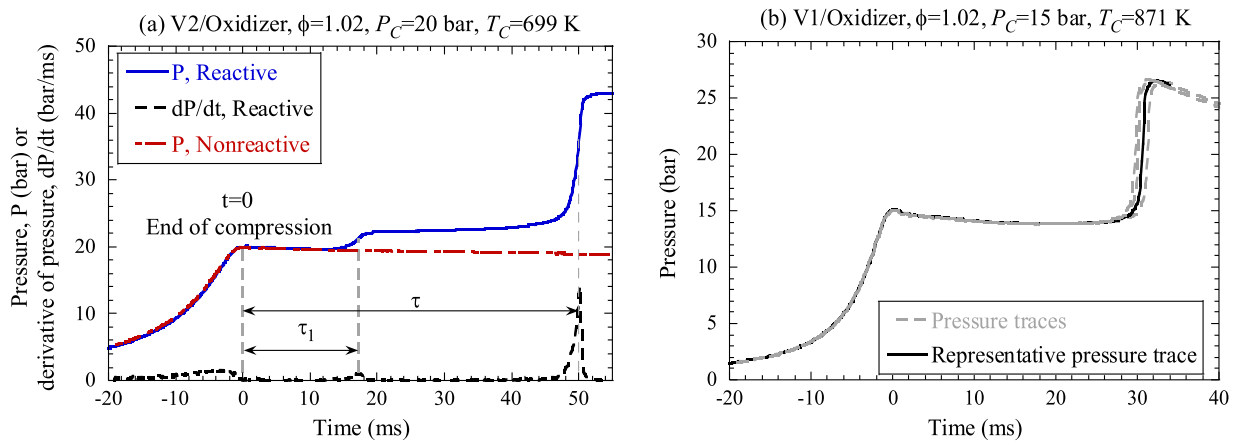


Fig. 4. (a) Definitions of first-stage and total ignition delay times, and the corresponding nonreactive pressure trace of V2/oxidizer at $P_c = 20$ bar, $T_c = 699$ K, and $\phi = 1.02$ with dilution. (b) Pressure traces of V1/oxidizer at $P_c = 15$ bar, $T_c = 871$ K, and $\phi = 1.02$ with dilution showing the experimental repeatability of the current study.

pressure histories as a function of time measured using the pressure transducer located at the manifold, which is heated to the set preheat temperature and monitored by a thermocouple. This pressure transducer covers the low pressure range and has a full-scale uncertainty of 0.05%, thus resulting in an error bar of ± 2.6 Torr shown in Fig. 3. Note that V0a and V0b are respectively the most and least volatile surrogates among the four CRC surrogates. In addition, the fuel masses injected into the preheated mixing tank

for the results shown in Fig. 3 are 1.64 g, 1.85 g, and 0.93 g for CFA, V0a, and V0b, respectively, which are equal to or even more than the mass of fuel in the highest fuel loading condition tested in the present RCM experiments. It is seen from Fig. 3 that for CFA and CRC surrogates, more than 85% of the reference pressure was reached instantly after the liquid fuel was injected into the tank. After that, the pressure reached $\sim 95\%$ of the expected value rapidly and this remained unchanged for more than an hour. These

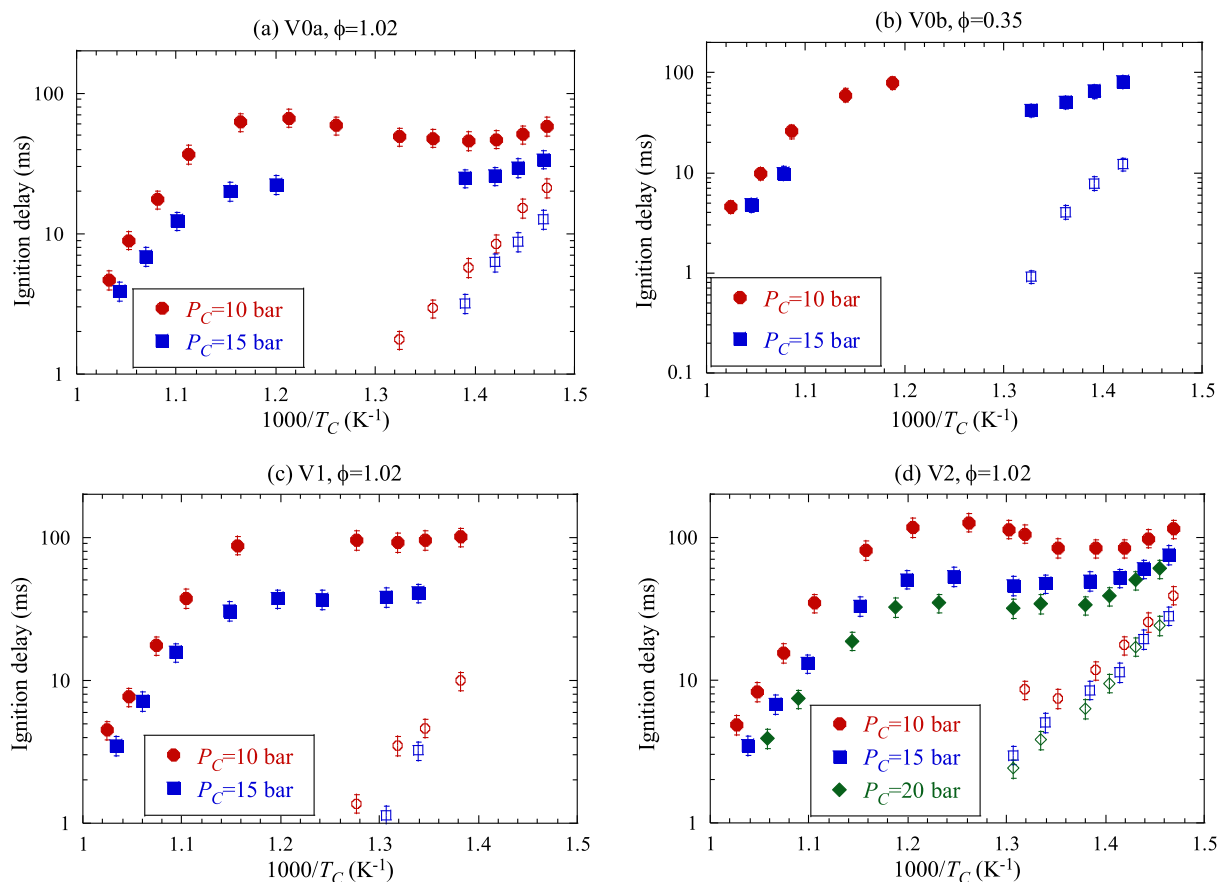


Fig. 5. Effect of compressed pressure on τ (filled symbols) and τ_1 (open symbols) for (a) V0a/oxidizer mixtures at $\phi=1.02$ with dilution, (b) V0b/oxidizer mixtures at $\phi=0.35$ in air, (c) V1/oxidizer mixtures at $\phi=1.02$ with dilution, and (d) V2/oxidizer mixtures at $\phi=1.02$ with dilution.

Table 6

Total ignition delay time scatter for CFA diesel and the four CRC surrogates at the representative conditions.

ϕ	P_C (bar)	T_C (K)	Scatter of CFA	Scatter of V0a	Scatter of V2	Scatter of V1	Scatter of V0b
0.5	10	702	3.1%	1.6%	7.9%	4.7%	2.2%
2.0	20	840	5.1%	8.2%	6.7%	3.0%	6.0%
2.0	20	920	4.2%	4.9%	8.3%	2.5%	1.2%

vaporization checks therefore demonstrate that the uncertainty of fuel partial pressure in the mixing tank is consistently ~5%.

2.3. Ignition delay time definitions and experimental repeatability

Figure 4(a) shows a representative pressure trace of V2/oxidizer at $P_C=20$ bar, $T_C=699$ K, and $\phi=1.02$ with dilution, demonstrating the definitions of ignition delay times used in this study. The end of compression point was regarded as the starting point of ignition delay time, which is set as $t = 0$ ms. In Fig. 4(a), τ_1 indicates the first-stage ignition delay time and τ indicates the total ignition delay time, with both of them being identified based on the respective local maximum of the time derivative of the pressure trace. The nonreactive pressure trace of the same experimental condition has also been shown in Fig. 4(a) as a reference. For each reactive set of P_C and T_C , the corresponding nonreactive pressure trace is measured by replacing oxygen with nitrogen to characterize the heat transfer effect on the ignition process and to verify that no heat release has occurred during the compression stroke. Four to five consecutive runs were taken for each experimental condition to ensure data repeatability, which can be seen in Fig. 4(b) with representative pressure traces of V1/oxidizer at $P_C=15$ bar, $T_C=871$ K, and $\phi=1.02$ with dilution. The typical scat-

ter was found to be less than 10% of the reported ignition delay time value, which is the closest to the mean of the consecutive runs. After taking the major uncertainties related to ignition delay time measurements into consideration, such as the uncertainties in mixture composition, measurements of initial pressure and temperature, and the facility-dependent non-ideal processes during and after compression, the current ignition delay time data of all four to five consecutive runs for each condition still fall into the 10% scatter range.

The scatter in total ignition delay time data of CFA diesel and the four CRC surrogates are compared in Table 6 to provide an additional check on potential fuel condensation issues. Scatter is defined here as the range in τ ($\pm\%$) of the consecutive runs based on the representative τ value. If condensation were to occur in experiments of surrogates containing more heavy components, the scatter in τ is expected to increase with increasing heavy component mole fraction in the surrogate due to increasing variability in the experimental results. NC18H38 is considered the heavy component of most concern for condensation in the surrogates. Based on the surrogate compositions reported in Mueller et al. [2], the mole fraction of NC18H38 in the surrogates increases in the order of V0a < V2 < V1 < V0b, where V0a has no NC18H38. If condensation

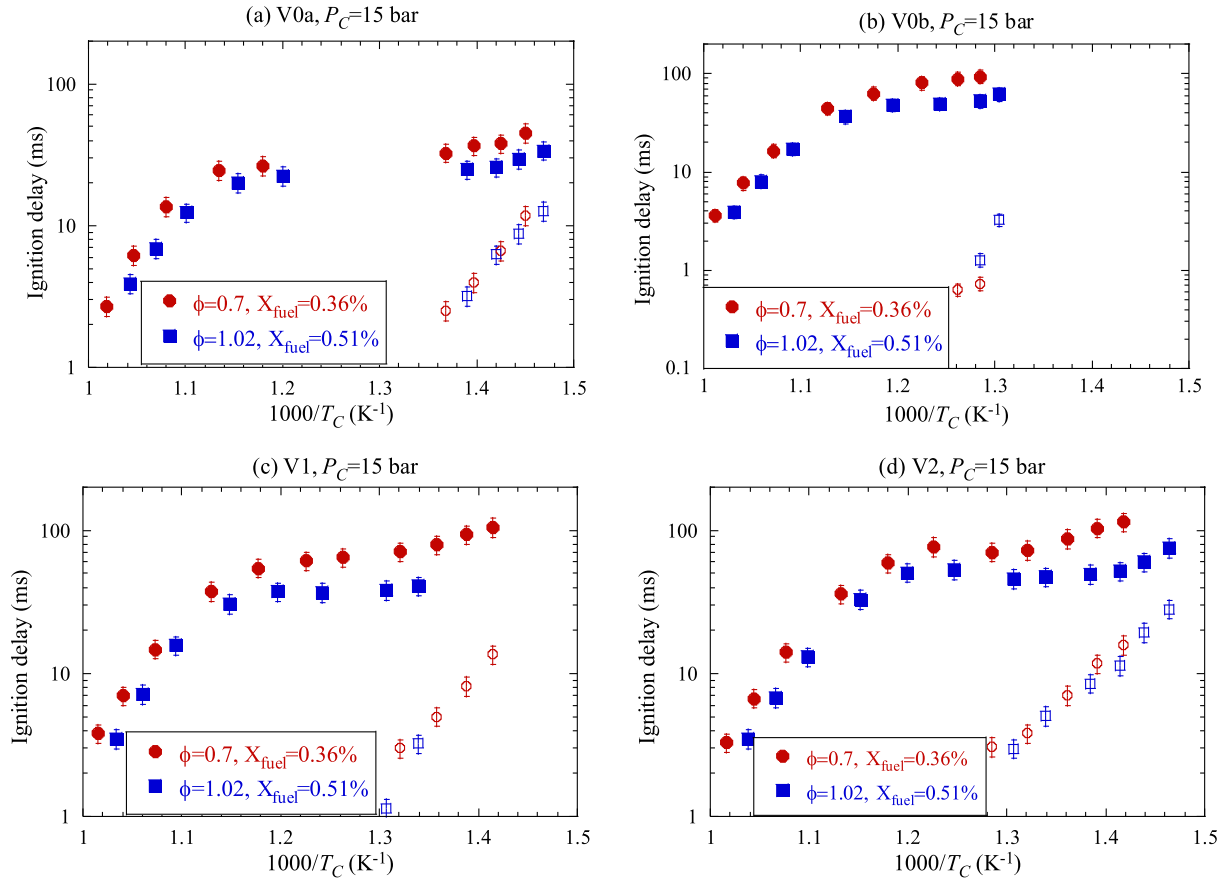


Fig. 6. Effect of fuel loading on τ (filled symbols) and τ_1 (open symbols) at $P_C=15$ bar for (a) V0a, (b) V0b, (c) V1, and (d) V2.

Table 7

Total ignition delay time scatter at varying temperatures for CFA diesel and the four CRC surrogates.

T_C (K)	$\varphi=1.0$, $P_C=20$ bar			$\varphi=0.69$, $P_C=10$ bar	$\varphi=0.5$, $P_C=10$ bar
	Scatter of CFA	Scatter of V0a	Scatter of V2	Scatter of V1	Scatter of V0b
702	4.6%	4.6%	4.8%	2.1%	2.2%
840	5.1%	8.2%	6.7%	8.7%	–
920	4.2%	4.9%	8.3%	4.5%	6.3%

were an issue, it is expected to introduce a pronounced variability in τ and the τ scatter would be expected to increase in this order. However, the τ scatter shown in Table 6 is not increasing in this order but is somewhat random. Table 7 further shows the τ scatter for CFA diesel and the four CRC surrogates at different temperatures representing LTR, NTC regime, and ITR. It is noted that the τ scatter of V0b at $T_C=840$ K is not shown in Table 7, which is because this data point at $\varphi=0.5$ and $P_C=10$ bar is not available due to some exothermicity during the compression stroke. Again, for CFA diesel and each surrogate, the scatter in τ is not decreasing as temperature increases as would be expected if condensation were occurring. Therefore, it can be inferred from both Table 6 and Table 7 that there is no apparent fuel condensation issue in the current RCM experiments.

3. Results and discussion

3.1. Effects of pressure, fuel loading, and oxygen concentration on ignition delay time

Figure 5 shows the effect of varying pressure on total and first-stage ignition delay times of the four CRC surrogates. It can

be observed that both τ and τ_1 decrease with increasing pressure, but the effect of pressure on τ_1 is less pronounced. In addition, the NTC slope becomes less pronounced as pressure increases. For V1 at $\varphi=1.02$, it is seen from Fig. 5(c) that only part of V1 data points in the LTR are reported. This is because for $P_C=10$ bar, the total ignition delay times in the LTR are more than 100 ms for data points at temperatures lower than 725 K. For $P_C=15$ bar, on the other hand, the total pressure in the mixing tank at 420 K preheat cannot provide sufficient initial pressure to reach the target P_C for data points at temperatures lower than 745 K, which is the limitation of partial pressure discussed earlier.

Figure 6 shows the effect of varying fuel loading on ignition delay times for the four CRC surrogates at $P_C=15$ bar. It can be observed that the total ignition delay times are seen to decrease with increasing equivalence ratio (fuel loading) from $\varphi=0.7$ to $\varphi=1.02$, but there is not much fuel loading effect observed on first-stage ignition delay times. The Arrhenius plot is also observed to shift to lower temperature with increasing fuel loading. Figure 7 further demonstrates the effect of oxygen concentration on ignition delay times of the four CRC surrogates at $P_C=10$ bar. It is seen that lowering the oxygen mole fraction increases both τ and τ_1 , and this

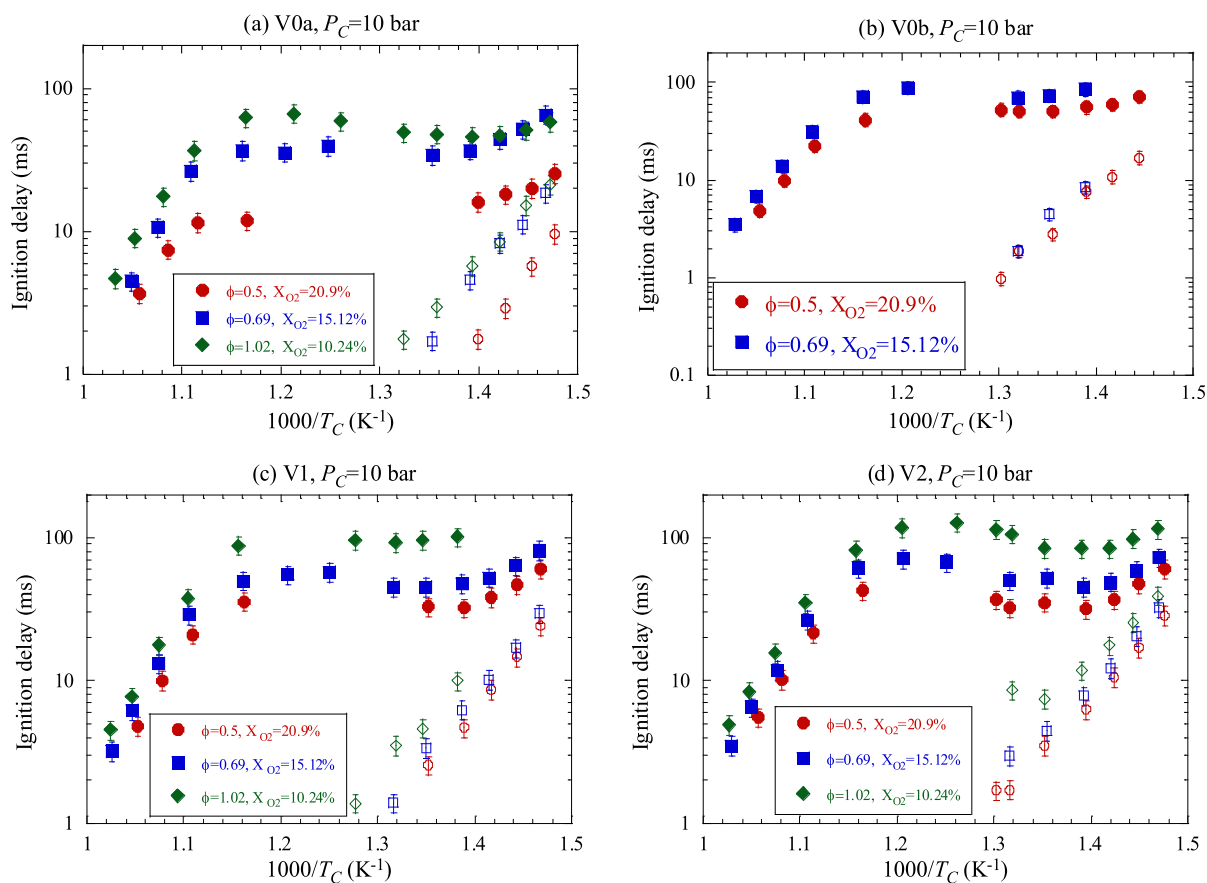


Fig. 7. Effect of oxygen concentration on τ (filled symbols) and τ_1 (open symbols) at $P_C=10$ bar for (a) V0a, (b) V0b, (c) V1, and (d) V2.

effect on total ignition delay time is still more pronounced, especially in the NTC regime.

3.2. Autoignition comparison of CRC surrogates and CFA

Figure 8 compares the ignition delay time results of the four CRC surrogates and CFA under various experimental conditions listed in Table 4. According to the current results, it can be concluded that the discrepancies of total ignition delay times of CFA and CRC surrogates are more noticeable in the LTR and NTC regime, while their total ignition delay times gradually overlap as temperature increases. In addition, the sequence of ignition delay times of CFA and its diesel surrogates is also in reasonable agreement with their DCNs [2] at temperatures of 790–850 K, which is the temperature range determining DCN by the ASTM standard test procedure, with V0a having the shortest total ignition delay times while V2 and V0b having the longest ones. For the first-stage ignition delay times, while the discrepancies are smaller compared to total ignition delay times in the LTR, similar ranking can still be observed, with the first-stage ignition delay times of V1, V2, and V0b consistently longer than those of CFA and V0a.

To further access how well the four CRC surrogates emulate the ignition response of CFA, their experimental pressure traces are compared at the same temperature (machine settings) in RCM experiments. Figure 9 demonstrates these comparisons for representative pressure traces with three different temperatures for $\phi = 0.5$ in air/ $P_C = 10$ bar, $\phi = 0.7$ with dilution/ $P_C = 15$ bar, and $\phi = 1.02$ with dilution/ $P_C = 15$ bar, respectively. Nonreactive pressure traces with the same test conditions and machine settings have also been shown in Fig. 9 to substantiate that there is no exothermicity in the compression stroke and to demonstrate the extent of heat loss

after the end of compression. It can be observed from Fig. 9(a) that with similar heat loss characteristics, the surrogate with a shorter total ignition delay time has a higher first-stage ignition pressure rise. In terms of the peak pressure after hot ignition, V0a is about 10–15% higher than CFA, while V0b, V1, and V2 are more comparable with CFA. However, it has been mentioned in Mueller et al. [2] that the net heat of combustion of V0a is about 1% higher than CFA and other surrogate fuels. This difference may be due to the multi-dimensional effects inside the reaction chamber that occur during the hot ignition process involving the exchange of mass and energy among the core region, the boundary layer, and the piston crevice [44]. These factors, coupled with the chemical reactions and heat release, could lead to different pressure and temperature rises after hot ignition for different CRC surrogates, with the most reactive V0a exhibiting the highest peak pressure.

4. Surrogate model development and simulated results

A detailed diesel surrogate model containing all components of V0a, V0b, and V1 has been developed by merging component models recently developed at Lawrence Livermore National Laboratory, component models from the literature, and the C_0 – C_4 base mechanism from AramcoMech 2.0 [45]. This diesel surrogate model has been included in Supplementary Material. All the sub-models used to generate the current surrogate model have been validated against literature data, and the mechanism in general was found to capture the ignition delay times of pure components (including *n*-decane, *n*-dodecane, isocetane, *trans*-decalin, tetralin, 1,2,4-trimethylbenzene, and 1-methylnaphthalene), as well as binary, ternary, and more complex surrogates with good level of agreement. As there are no literature studies on autoignition of

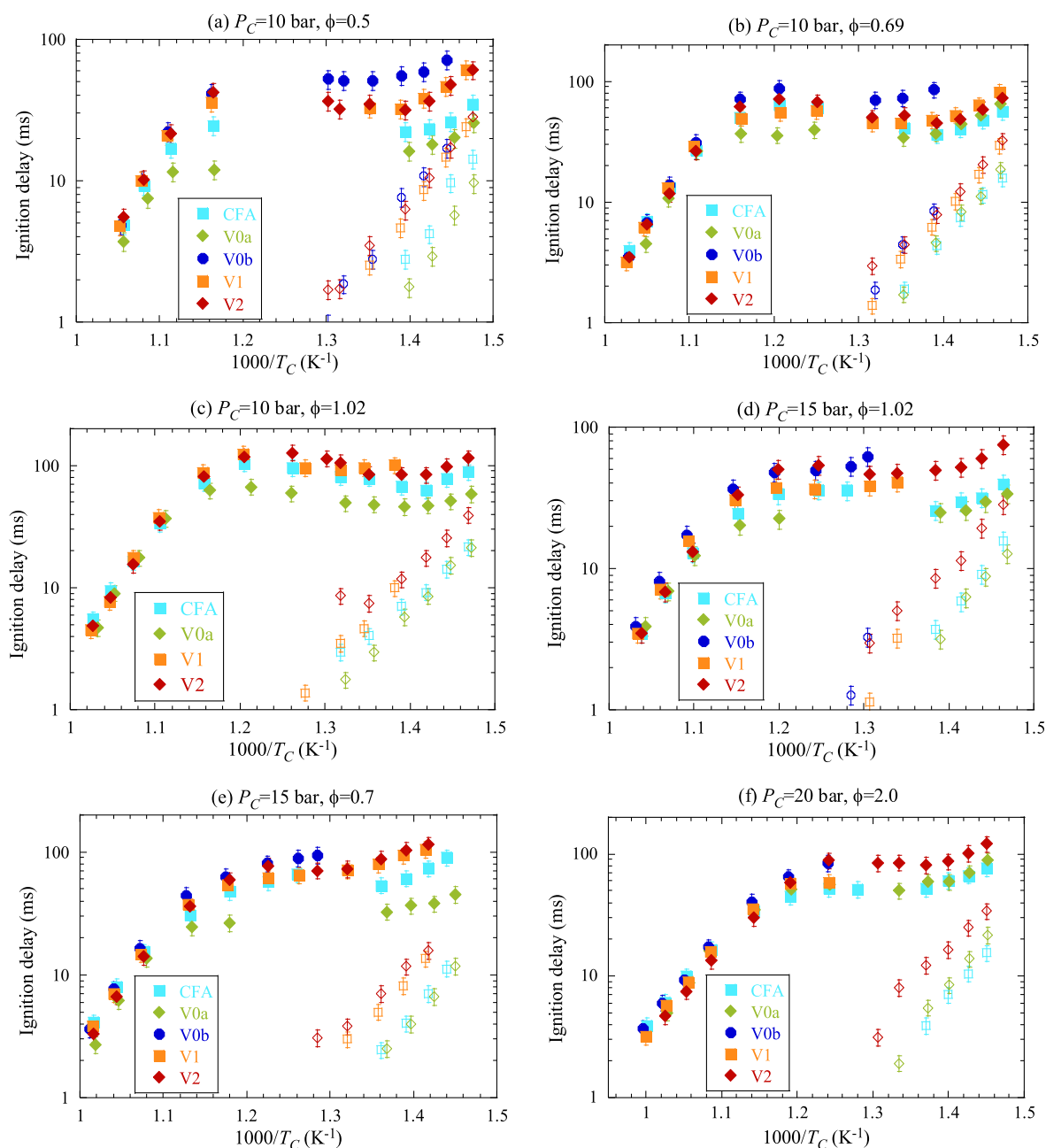


Fig. 8. Autoignition results comparison of CRC surrogates and CFA at all test conditions shown in Table 4. Filled symbols correspond to total ignition delay times and open symbols correspond to first-stage ignition delay times.

n-octadecane and *n*-eicosane, to the best of our knowledge, their sub-models were developed using consistent reaction classes and rate rules of smaller *n*-alkanes because they are homologous hydrocarbons and are expected to have analogous oxidation chemistry and similar ignition characteristics. More details about the model development and validation have been demonstrated in Guzman et al. [46] and Kukkadapu et al. [47].

RCM simulations have been performed using the current detailed diesel surrogate model in Chemkin-Pro [48]. The heat transfer effect has been considered by using volume traces derived from nonreactive experiments; the details of which have been described previously in [42]. The predictions by the current surrogate model are shown in Fig. 10 with lines. The overall agreement between the experimental and simulated ignition delay times is reasonable,

and the model successfully predict the overall ranking of both total and first-stage ignition delay times for V0a, V0b, and V1 surrogates. However, it appears that the reactivity of both first-stage and total ignition delay times in the LTR have been over-predicted. Furthermore, although the simulated results show that the total ignition delay times of all surrogates gradually overlap in the ITR, which is the same trend in experimental results, the model predicts longer total ignition delay times than experimental data, suggesting that further refinements are required to improve the performance of the current surrogate model. It can also be noticed that the simulated results of certain experimental conditions have not been shown in Fig. 10, which is because exothermicity during the compression stroke is noted in simulated pressure traces by comparing with the corresponding nonreactive experimental

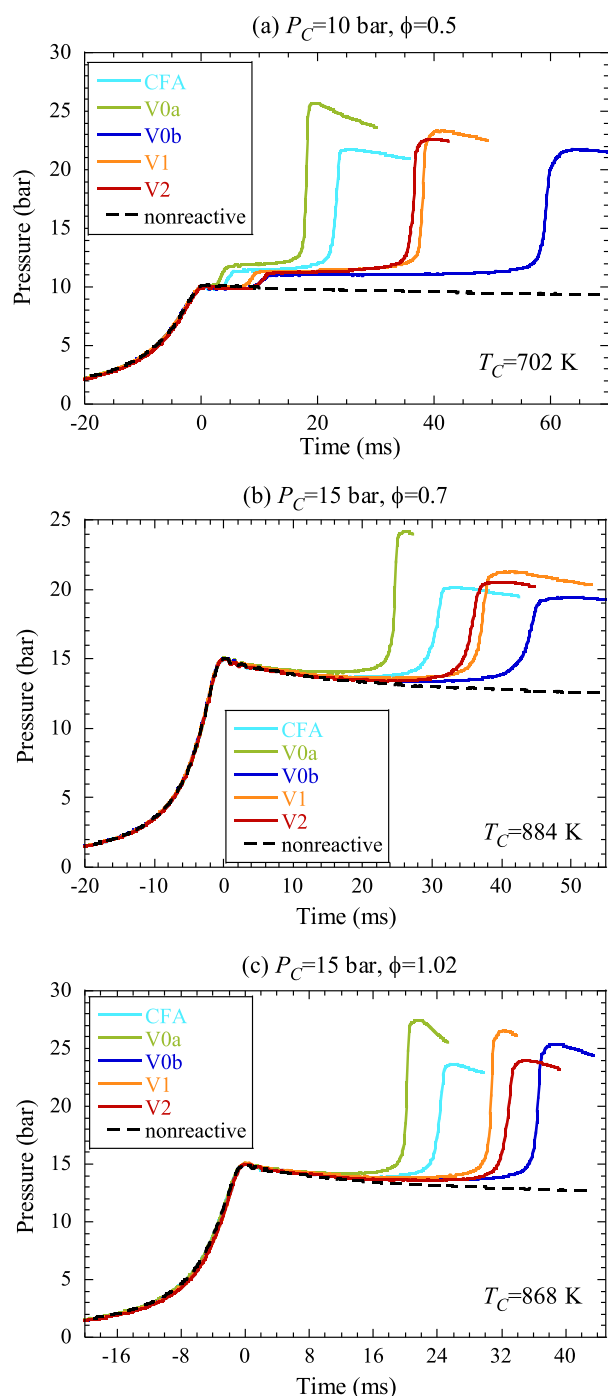


Fig. 9. Plot showing the pressure trace comparison of CFA and CRC surrogates under test conditions of (a) $\phi=0.5$ in air/ $P_C=10$ bar, (b) $\phi=0.7$ with dilution/ $P_C=15$ bar, and (c) $\phi=1.02$ with dilution/ $P_C=15$ bar.

pressure traces with similar initial pressure. An example demonstrating exothermicity in simulation during the compression stroke is shown in Fig. S1(b) in SM, with exothermicity marked by a red rectangle. Further refinements in the model are needed to address the discrepancies at these conditions.

Figure 11 compares the experimental pressure traces with simulated results of V0a, V0b, and V1 surrogates at two temperatures for $\phi = 0.69$ with dilution/ $P_C = 10$ bar and $\phi=1.02$ with dilution/ $P_C=15$ bar, representing LTR and ITR, respectively, to get a better understanding of the discrepancies between experimen-

tal and simulated results. It can be observed from Fig. 11(a) that the model predicts higher first-stage pressure rises as compared to the experimental results for all CRC surrogates, which may be due to the multi-dimensional effects of RCM experiments occurring after the first-stage ignition event are not fully captured in the zero-dimensional simulation. In addition, the higher simulated first-stage pressure rise may lead to shorter second-stage ignition delay time compared to the experimental data. Furthermore, it can be observed from both Fig. 11(a) and (b) that the most reactive V0a also has the highest simulated peak pressures after hot ignition, which is consistent with the experimental results. This may indicate that other than multi-dimensional effects, chemical reactions and heat release could also lead to different pressure rises for different surrogates, as discussed earlier.

In order to further understand the discrepancies between experimental and simulated results, brute force sensitivity analyses to identify the controlling chemistry for autoignition have been performed under constant volume adiabatic (CONV) simulations based on the current diesel surrogate model. These analyses are conducted at initial pressure of 10 bar, $\phi=0.69$ with dilution, under initial temperatures of 700 K, 800 K, and 950 K (representing LTR, NTC, and ITR, respectively) for V0a, V0b, and V1 surrogates to highlight the important reactions that affect the predicted reactivity for surrogates. For simplicity, τ (τ_1) in the present CONV simulations is defined as the time at which the local temperature is 400 K (10 K) above the initial temperature in order to facilitate sensitivity analyses. The ignition delay times obtained using the temperature rise definition are within 1 ms from those determined by the local maximum of time derivative of pressure trace, as demonstrated in Fig. S2 in Supplementary Material using CONV simulation of V1 at $\phi=0.69$ with dilution under initial temperature of 700 K and initial pressure of 10 bar as an example. The relative sensitivity (S_{rel}) is defined as $S_{rel} = \ln(\tau_+/ \tau) / \ln(k_+/k)$, where k and τ are the unperturbed reaction rate constant and the corresponding original total ignition delay time, respectively, and k_+ is the rate constant increased by a factor of two while τ_+ is the resulting total ignition delay time. The sensitivity analyses for first-stage ignition delay time are also carried out in the same manner. Therefore, a reaction with a positive sensitivity coefficient inhibits reactivity while that with a negative sensitivity coefficient promotes reactivity.

For the total ignition delay time sensitivity analysis of V0a, Fig. 12(a) shows the top ranked most sensitive reactions. It is seen that the H-abstraction reactions of the surrogate components with OH radical and O_2 play an important role in controlling the total ignition delay time at 700 K. All four components of V0a are included in H-abstraction reactions shown in Fig. 12(a), and the sensitivity coefficients of these reactions exhibit site specificity. Specifically, the H-abstraction reactions for HMN and DECALIN leading to the tertiary iso-cetyl (HMN-R5) and tertiary decalyl (DCL-1R) radicals exhibit positive sensitivity, thereby inhibiting the reactivity. This is because the radicals produced on those sites cannot further produce ketohydroperoxides (KHPs) through the subsequent propagation reactions, as no hydrogen remains on these sites for the second inner hydrogen transfer. The same reason can be applied to the H-abstraction reaction for A2CH3 producing methyl naphthalenyl radicals (C6H3A1CH3) which also exhibits positive sensitivity. In addition, the chain initiation reaction between A2CH3 and O_2 leading to the formation of 1-naphthylmethyl radical (A2CH2) and hydroperoxyl radical (HO_2) retards ignition. As discussed in Kukkadapu and Sung [49], this reaction proceeds in the reverse direction, thus competing with the reaction leading to the formation of naphthalene-1-carbaldehyde (A2CHO) and OH radicals that accelerates ignition as indicated by its negative sensitivity coefficient. Furthermore, the reactions that facilitate low temperature chain branching reactions such as first and second O_2 addition

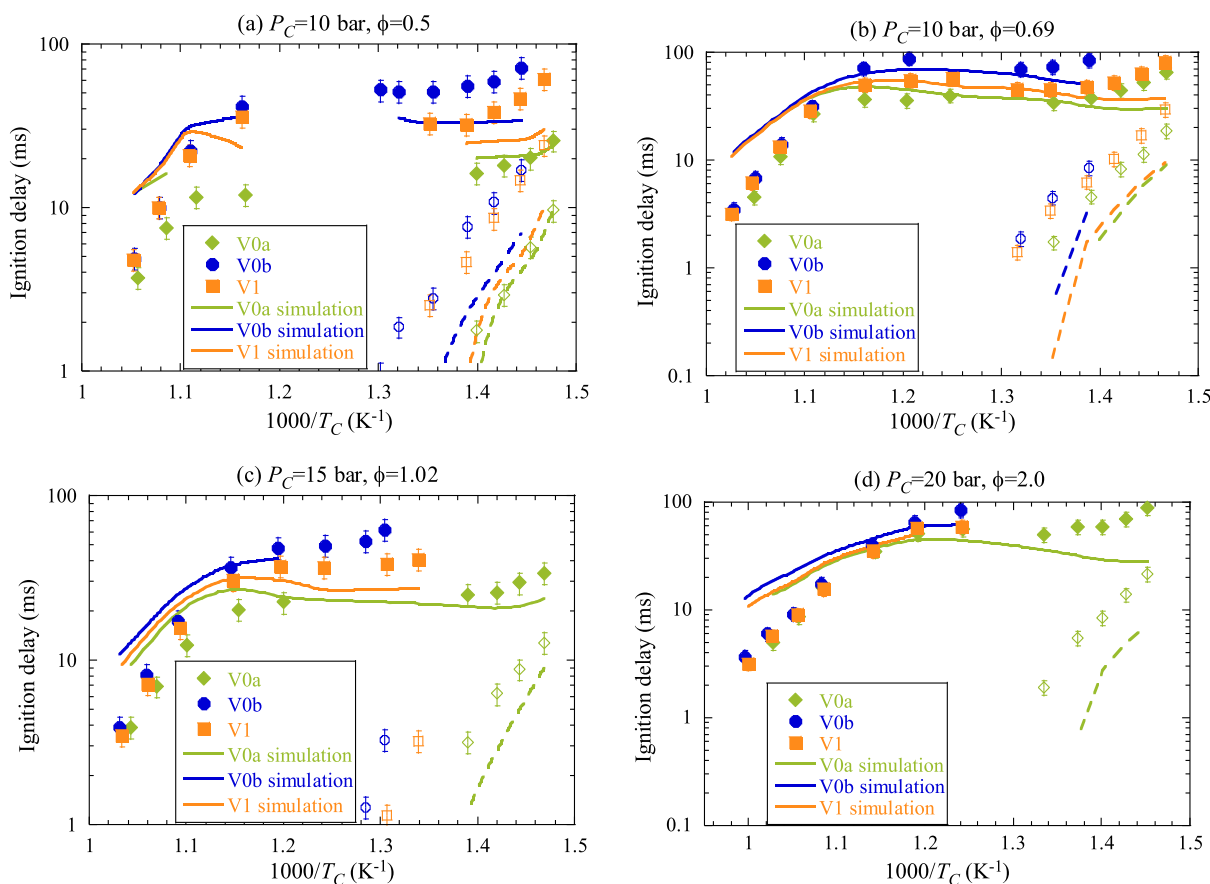


Fig. 10. Autoignition results comparison of experiments and simulations for CRC surrogates V0a, V0b, and V1 at (a) $\phi=0.5$ in air/ $P_c=10$ bar, (b) $\phi=0.69$ with dilution/ $P_c=10$ bar, (c) $\phi=1.02$ with dilution/ $P_c=15$ bar, and (d) $\phi=2$ with dilution/ $P_c=20$ bar. Filled symbols: total ignition delay times from experiments. Open symbols: first-stage ignition delay times from experiments. Solid lines: total ignition delay times from simulations using the current surrogate model. Broken lines: first-stage ignition delay times from simulations using the current surrogate model.

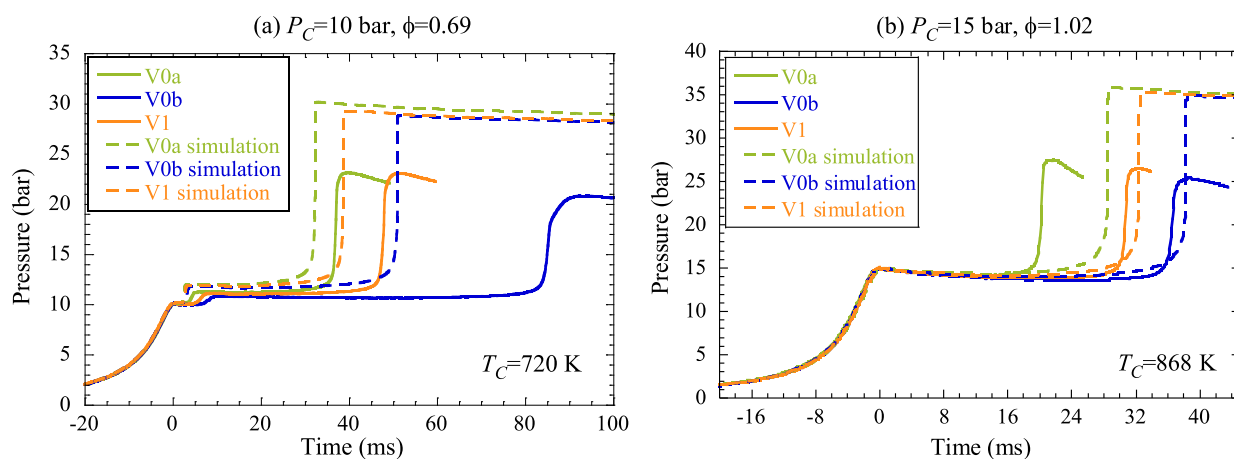


Fig. 11. Plot showing the experimental and simulated pressure trace comparison of CRC surrogates under test conditions of (a) $\phi=0.69$ with dilution/ $P_c=10$ bar and (b) $\phi=1.02$ with dilution/ $P_c=15$ bar.

reactions, isomerization of alkyl peroxides, and formation of KHPs, promote the reactivity, while other reactions such as concerted elimination of alkyl peroxides and formation of cyclic ethers inhibit the reactivity, which are typical chemical behaviors at low temperatures.

It is also shown in Fig. 12(a) that similar classes of reactions govern the total ignition delay time as initial temperature is increased to 800 K. Moreover, for the reactions that have positive

sensitivity coefficients, H-abstraction reactions of parent fuels HMN and A2CH3 forming HMN-R5 and A2CH2, respectively, as well as concerted elimination reactions and cyclic ethers formation reactions become more pronounced when temperature approaching the NTC regime. Both the formation (inhibiting) and decomposition (promoting) reactions of H_2O_2 in Fig. 12(a) also become more dominant, thus controlling the predicted reactivity in the NTC regime.

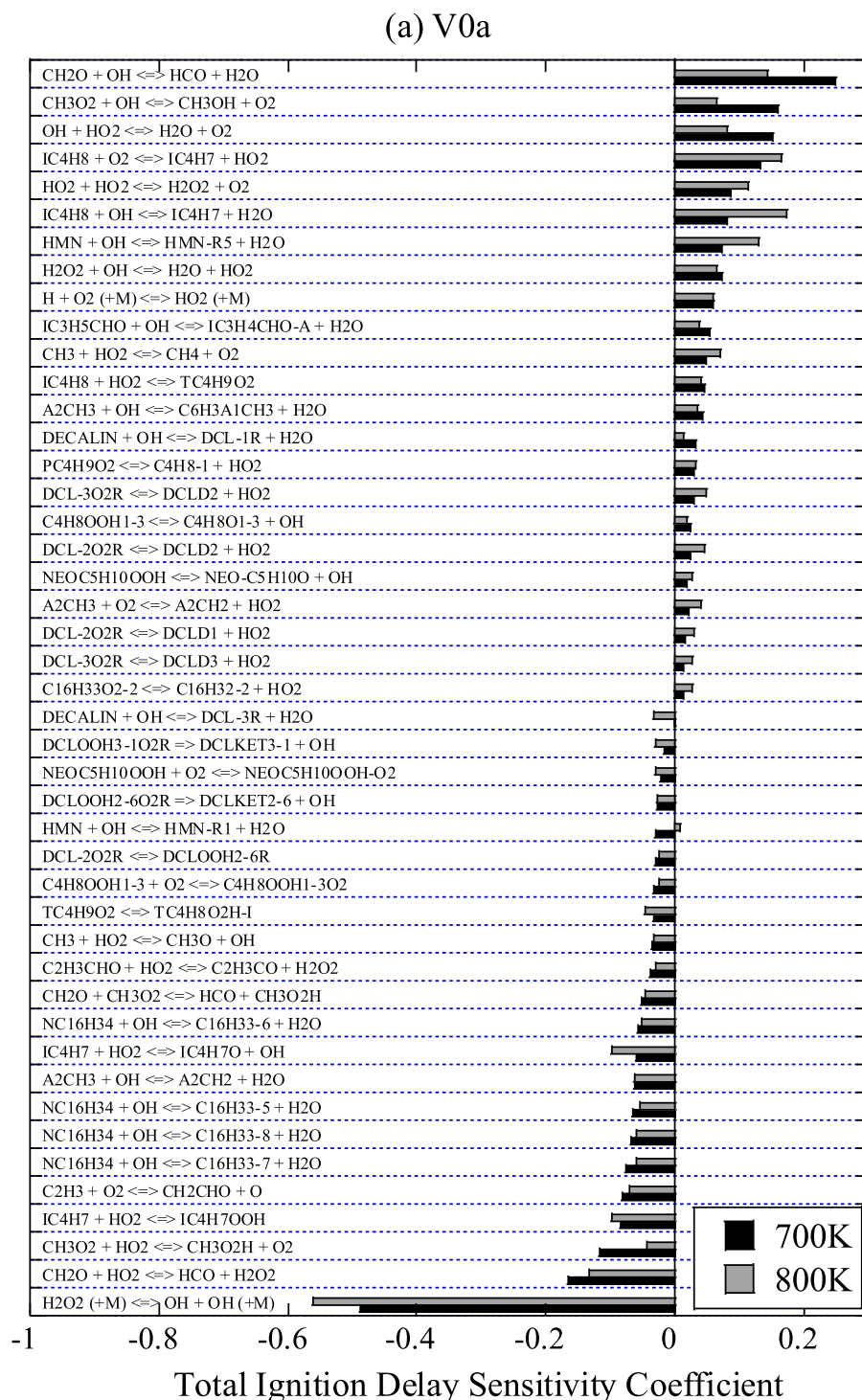


Fig. 12. Brute force sensitivity coefficients for total ignition delay times of V0a, V0b, and V1 surrogates at $\phi = 0.69$ with dilution and initial pressure of 10 bar with varying initial temperatures.

Figure 13(a) further shows the first-stage ignition delay time sensitivity analysis results of V0a at 700 K. It can be observed that the H-abstraction reactions of NC16H34 are the most dominant reactions promoting the reactivity of first-stage ignition, while the H-abstraction reactions of DECALIN and A2CH3 by OH radical become dominant inhibiting reactions. A possible explanation is that *n*-alkanes are the first to be consumed compared to other hydrocarbon classes during low temperature ox-

idation, thereby producing OH radicals that can further help the consumption of other surrogate fuel components, which will be demonstrated later in detail. It is also noted that the isomerization reaction of HMN-R5O2 forming HMNOOH5-2 is an important reaction promoting the first-stage ignition reactivity, while it is not among top reactions promoting the reactivity of total ignition. Conceivably, this is because most HMNOOH5-2 then form cyclic ethers and produce OH radicals, and these OH radicals can help promote

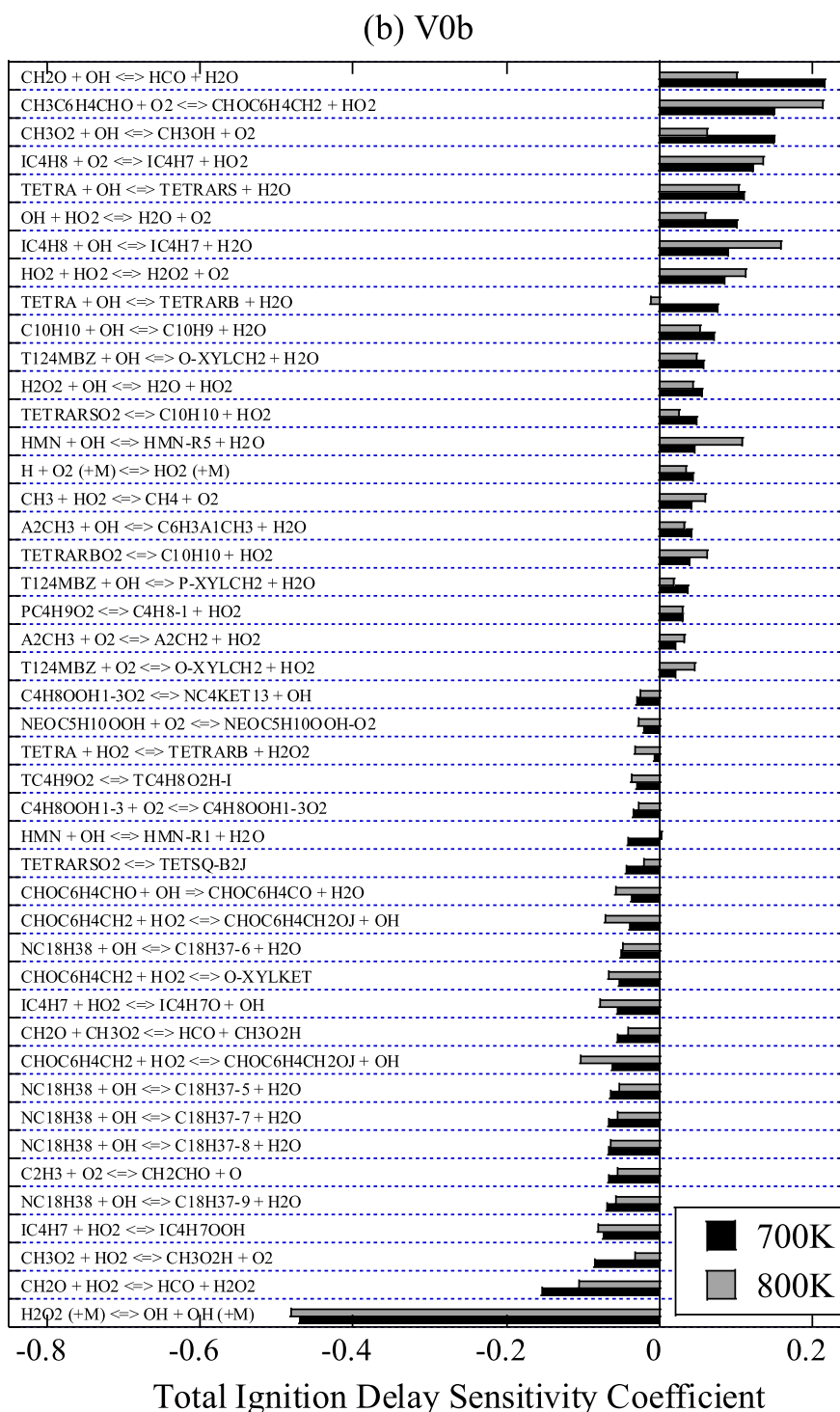


Fig. 12. Continued

the first-stage ignition. Based on the discrepancies between experimental and simulated results of V0a shown in Fig. 10, both τ in the LTR and τ_1 are under-predicted, while the agreement with experimental results is better in the NTC regime. In addition, based on the dominant reactions in the sensitivity analysis results of V0a, it is suggested that various sensitive reactions need to be refined in concert in order to improve the LTR ignition delay time prediction without affecting the performance in the NTC regime. One possible way for better model prediction is to conduct experimental autoignition studies of NC16H₃₄, HMN, and/or their blends at low

temperatures, not only because they are important in controlling the ignition delay times, but also the autoignition characteristics of these fuels at low temperatures are not well understood [50–54], which may be a main source of discrepancy to experimental data. Further model development can then be conducted after obtaining the needed validation datasets.

Figure 12(b) and (c) demonstrates the top ranked most sensitive reactions for total ignition delay times of V0b and V1 at initial temperatures of 700 K and 800 K, and it can be observed that the fuel-component-related reactions controlling the

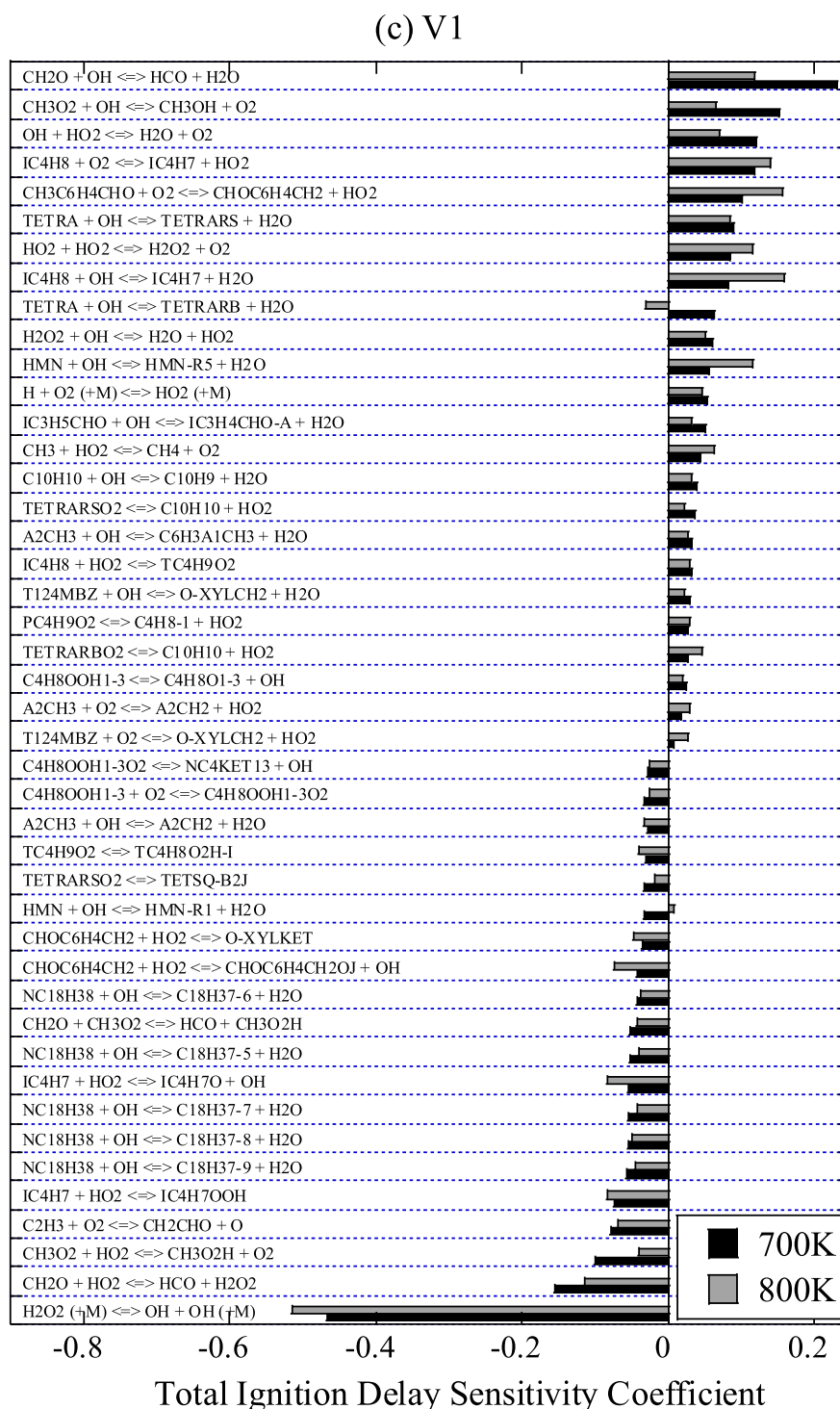


Fig. 12. Continued

reactivity are the same for both V0b and V1. Specifically, H-abstraction of NC18H38 by OH, as well as the H-abstraction reactions of HMN and A2CH3 leading to the primary iso-cetyl (HMN-R1) and A2CH2 radicals, promote the reactivity. The isomerization reaction of TETRARSO2 (RO₂ radical, where O₂ attached to the secondary carbon on the benzylic ring) producing TETSQ-B2J (QOOH radical) is another dominant reaction related to the fuel components that V0b and V1 have in common. Dominant reactions inhibiting the reactivity include H-abstraction of TETRA and T124MBZ. For TETRA, it is because, as discussed in Kukkadapu et al.

[55], that most of the tetralyl radicals formed from H-abstraction reactions by OH and O₂ then react with O₂ to form tetralyl peroxy, and the majority of the formed tetralyl peroxy subsequently participate in the concerted elimination reactions forming dialin (C10H10) and hydroperoxyl radicals (HO₂), which are also important reactions inhibiting the reactivity. For T124MBZ, it is because there are no hydrogen remain for the second inner hydrogen transfer for the radicals to further produce KHPs, as mentioned earlier. Moreover, the H-abstraction reactions of HMN and A2CH3 by OH radicals leading to HMN-R5 and C6H3A1CH3 and the H-abstraction

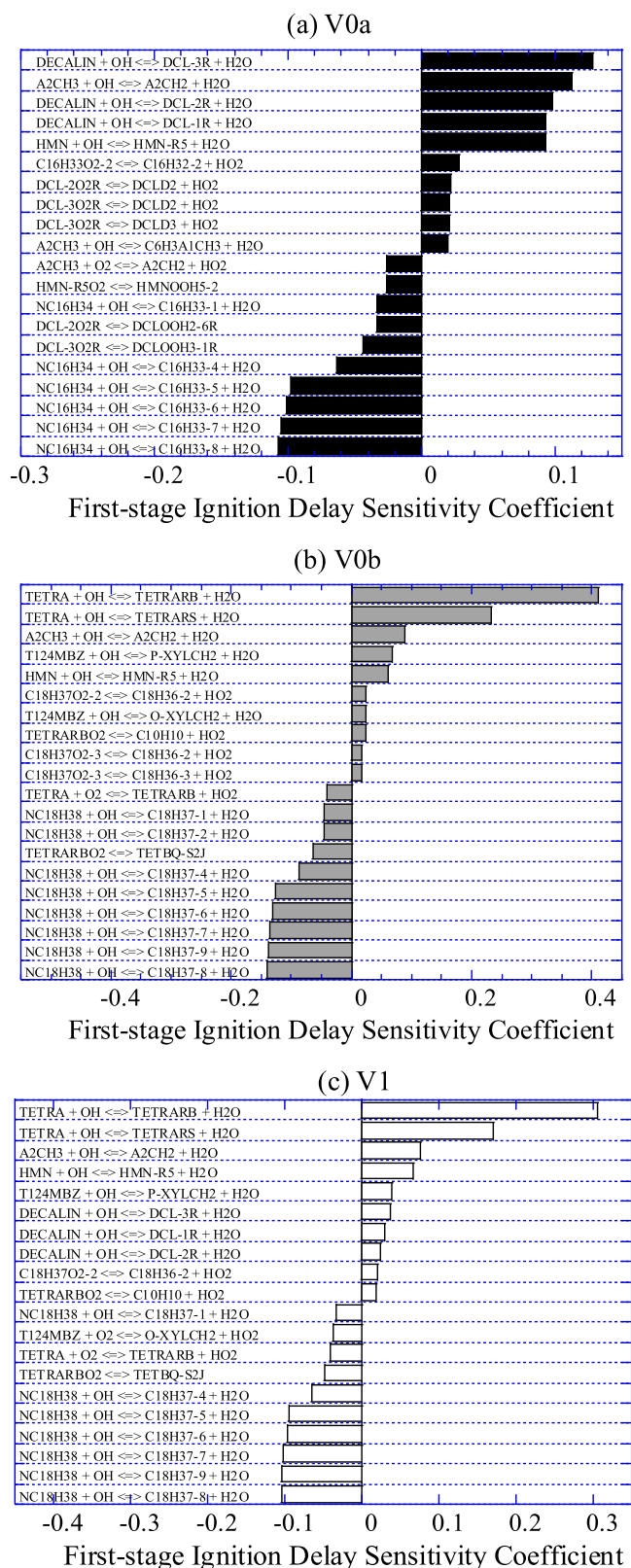


Fig. 13. Brute force sensitivity coefficients for first-stage ignition delay times of V0a, V0b, and V1 surrogates at $\phi=0.69$ with dilution and initial pressure/temperature of 10 bar/700 K.

reaction of A2CH3 by O₂ forming A2CH2 both inhibit the reactivity, as illustrated earlier in the dominant inhibiting reactions of V0a. The chain termination reaction between HO₂ radicals producing H₂O₂ and O₂ and the decomposition of H₂O₂ are still important in promoting and inhibiting reactivity, and they also become more dominant as initial temperature increases to 800 K, thus controlling the predicted reactivity in the NTC regime. The second O₂ addition reactions to QOOH radicals and the KHP formation reactions of intermediate species are other types of reactions promoting the reactivity, and other dominant reactions of intermediate species inhibiting the reactivity include the formation of cyclic ethers, which are also typical chemical behaviors at low temperatures and have been discussed previously.

As mentioned in the study of Mueller et al. [2], V0a and V0b are composed of simpler palettes which are subsets of the palette for the V1 surrogate (cf. Table 2). Based on the sensitivity analysis results on total ignition delay times of V0a, V0b, and V1, it can be observed that most dominant reactions in V1 can also be found in V0b, while the fuel-component-related reactions are more dominant in V0b reactivity than those of V0a are. Note that the components representing *n*-alkane in V0a and V0b are NC16H34 and NC18H38, respectively, which is likely the cause leading to the different sensitivities of the fuel-component-related reactions. In addition, DECALIN is no longer among top dominating reactions in total ignition sensitivity analysis results of V1. Apart from different reaction pathways as a result of different fuel components, another reason that the reactions involving NC16H34 and DECALIN and their radicals are not the controlling ones in V1 reactivity could be because their mole fractions in V1 are relatively lower than those in V0a.

The first-stage ignition sensitivity analysis results of V0b and V1 at initial temperature of 700 K are shown in Fig. 13(b) and (c), respectively. It is of interest to note the predominance of those sensitive reactions involving fuel-related species as compared to the total ignition sensitivity analysis results that mostly involve smaller intermediate species. Also, it is seen that most reactions controlling the reactivity of first-stage ignition are similar between V0b and V1, and some of these reactions are also dominant in controlling the total ignition reactivity, including the H-abstraction reactions of NC18H38, HMN, TETRA, A2CH3, and T124MBZ, as well as the concerted elimination reactions of alkyl peroxy radicals from NC18H38 and TETRA. It is of interest to note that the H-abstraction reactions of DECALIN are important reactions inhibiting the first-stage ignition reactivity in V1, while they are not among the top reactions controlling the total ignition reactivity of V1. As it can be observed from Fig. 10 that τ and τ_1 of V0b and V1 are under-predicted in the LTR, while the current surrogate model yields better agreement with the experimental results in the NTC regime, various sensitive reactions should be revisited in concert to achieve overall agreements in different temperature regimes. Furthermore, both the first-stage and total ignition sensitivity analysis results of V0b and V1 show that the H-abstraction reactions of NC18H38 and TETRA are the most dominant reactions in promoting and inhibiting the reactivity, respectively, among the reactions involving fuel components. Therefore, in order to further refine the current chemical kinetic model for improving its predictability, blends of NC18H38 and TETRA with varying blending ratios need to be studied for a better understanding of their chemical kinetic interactions. In addition, since the studies of [56–58] showed that including cross reactions between iso-alkanes and aromatics may have some influence on improving the model prediction, considering the possible cross reactions between HMN/TETRA and their radicals may improve the model performance. Besides that, autoignition of NC18H38, HMN, and/or their blends at low temperatures may also require further investigation, as the studies involving these fuels at low temperatures are meager.

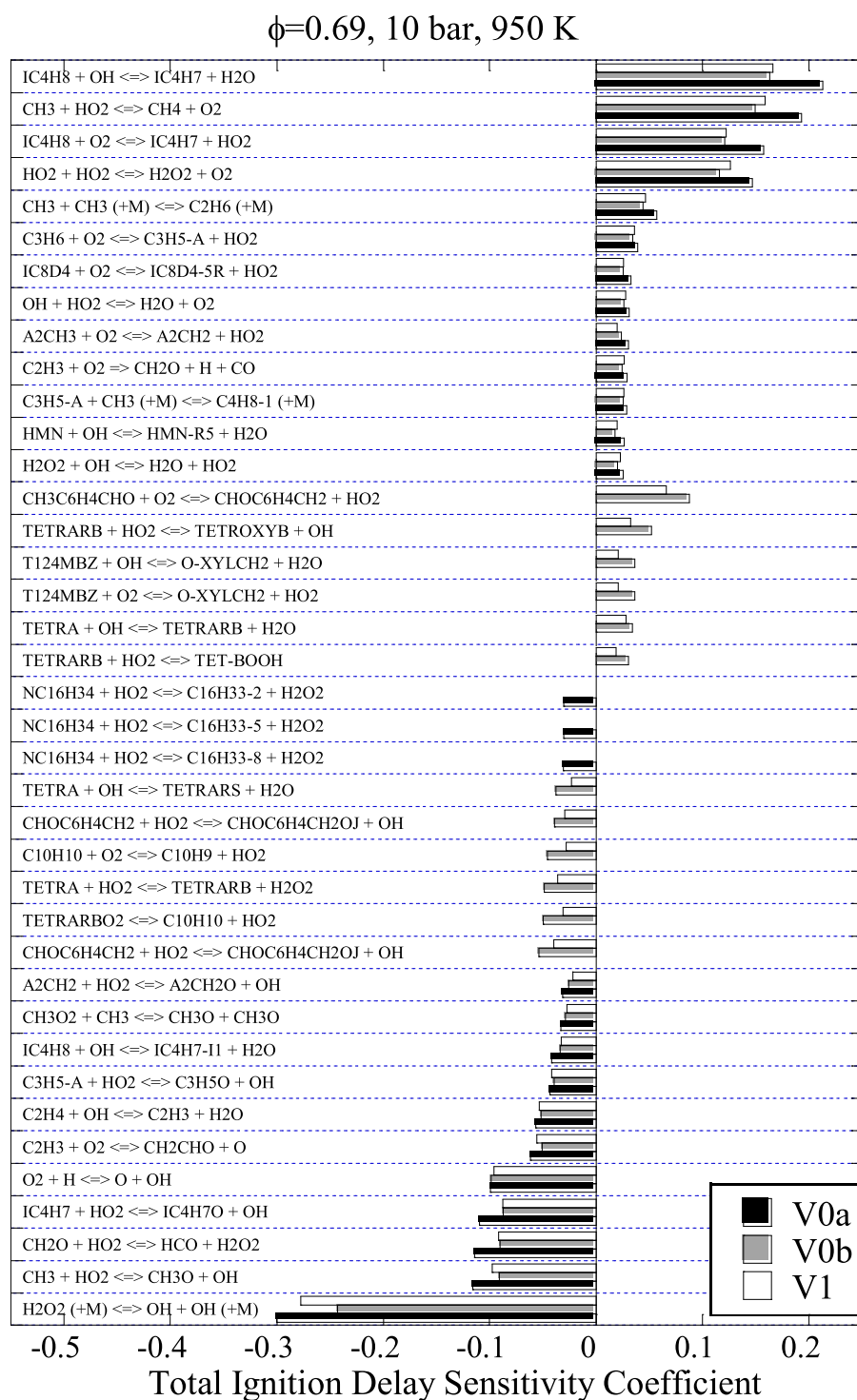


Fig. 14. Brute force sensitivity coefficients for total ignition delay times of V0a, V0b, and V1 surrogates at $\phi=0.69$ with dilution and initial pressure/temperature of 10 bar/950 K.

Figures S3 and S4 in Supplementary Material further show the total and first-stage ignition sensitivity analysis results of V0a, V0b, and V1 at other equivalence ratios and initial pressures. The comparison of these additional results with those shown in Figs. 12 and 13 for each CRC surrogate demonstrates that the controlling chemistry remains similar under varying conditions investigated, even at fuel rich condition with higher initial pressure.

Figure 14 shows and compares the top ranked most sensitive reactions for total ignition delay times of V0a, V0b, and V1 surro-

gates at $\phi=0.69$ with dilution, initial pressure of 10 bar, and initial temperature of 950 K, representing ITR. It can be observed that the most dominant reactions controlling the reactivity at 950 K are among smaller intermediate species, with chain branching reactions promoting ignition and chain terminating reactions inhibiting reactivity. In addition, Fig. 14 illustrates that the H-abstraction reactions of fuel components by HO₂ become dominant in this temperature regime, indicating that the concerted elimination reactions producing HO₂ are important reactions in the NTC regime.

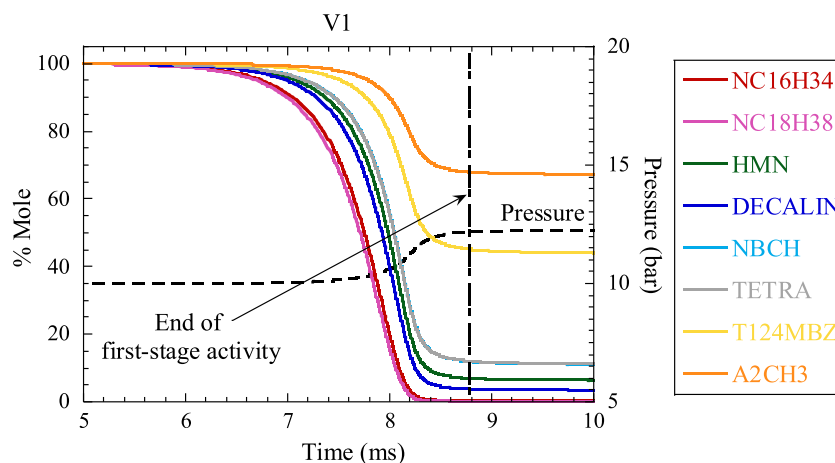


Fig. 15. Mole percent history of each fuel component in V1 surrogate along with simulated pressure trace as a function of time at $\phi=0.69$ with dilution and initial pressure/temperature of 10 bar/700 K.

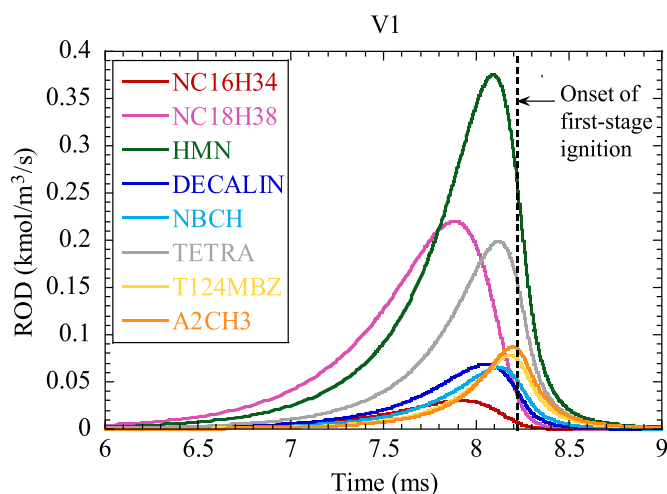


Fig. 16. ROD of each fuel component in V1 surrogate as a function of time at $\phi = 0.69$ with dilution and initial pressure/temperature of 10 bar/700 K.

Since the current model over-predicts total ignition delay times in the ITR, as shown in Fig. 10, the improvement in the model prediction at higher temperatures can be attributed to the refinement of the C_0 – C_4 base chemistry (i.e., AramcoMech2.0 [45]). Furthermore, the current surrogate model may also have better performance in different temperature regimes by improving the C_0 – C_4 base chemistry.

As mentioned previously, when exhibiting two-stage ignition response, both first-stage and total ignition delay times are under-predicted by the current diesel surrogate model (cf. Fig. 10), and better overall agreement with experimental data can be achieved by improving the prediction of first-stage ignition delay times. Therefore, chemical kinetic analyses have been conducted to understand the process of each fuel component being consumed before and during the first-stage ignition. Figure 15 shows an example of the mole percentage of each fuel component in V1 being consumed as a function of time at $\phi=0.69$ with dilution, initial pressure of 10 bar, and initial temperature of 700 K. Figure 16 further illustrates the rate of destruction (ROD) of each fuel component in V1 at the same test condition of Fig. 15. Not only can we observe the mole percentage of each fuel component residue at the end of first-stage activity from Fig. 15, but we can also see the sequence of each component being consumed as a function of time, which indicates and ranks the reactivity of those fuel com-

ponents in V1. This sequence can also be quantified by consumption time, which is the time to reach the maximum derivative of mole percentage consumption curve in Fig. 15, and this is also the corresponding time to reach the peak of ROD in Fig. 16. The mole percentage histories and ROD profiles of fuel components in V0a and V0b are shown in Figs. S5 and S6 in Supplementary Material, demonstrating that the consumption sequence of fuel components included in V0a and V0b is the same as that in V1.

From both Figs. 15 and 16, it is not surprising to observe that normal alkanes (NC16H34 and NC18H38) have the highest reactivity, while aromatics (T124MBZ and A2CH3) are the last to be consumed. Moreover, TETRA and HMN have been consumed almost simultaneously so that the cross reactions, if exist, can happen among them and their radicals. Hence, the chemical kinetic interactions between TETRA and HMN (not considered in the current model) can be further studied, as mentioned earlier. In Fig. 16, the time of the onset of first-stage ignition is shown as a dashed line. It is seen that the consumption times of fuel components included in hydrocarbon classes of normal alkanes, branched alkanes, cycloalkanes, and naphthoaromatics are all shorter than the first-stage ignition delay time, while aromatics reach their highest RODs almost simultaneously with the first-stage ignition. A possible explanation is that both T124MBZ and A2CH3 need radicals (i.e. OH) produced from low-temperature oxidation of other fuel components in order to initiate their consumption reactions.

5. Conclusions

Autoignition of the four CRC diesel surrogates have been studied in a rapid compression machine at low-to-intermediate temperatures and elevated pressures. The effects of pressure, fuel loading, and oxygen concentration on total and first-stage ignition delay times were demonstrated for the four CRC surrogates. In addition, the ignition delay times of the four CRC surrogates have been compared with those of CFA under various experimental conditions, with the discrepancies of total ignition delay times more noticeable in the LTR and NTC regime, and the total ignition delay times for all the fuels gradually overlap as temperature increases. The overall ranking of total ignition delay times for CFA and CRC surrogates is consistent with the ranking of their DCNs, especially in the temperature range for determining DCNs by the ASTM standard test procedure, and similar ranking can also be observed in first-stage ignition delay times. Furthermore, the experimental ignition delay times of CRC surrogates were used to assess the performance of a newly-developed chemical kinetic model including

all components of V0a, V0b, and V1. The agreement of the current model is reasonable in predicting the overall ignition characteristics, but further refinements are still needed for a better agreement, especially for total ignition delay times in the LTR and first-stage ignition delay times. Brute force sensitivity analyses were conducted at various pressures, temperatures, and equivalence ratios to identify the important reactions governing autoignition of V0a, V0b, and V1. In addition, the consumption process of each fuel component in those surrogates before and during first-stage ignition has also been analyzed in order to provide insights into further studies for model refinement, which include considering chemical kinetic interactions of NC18H38/TETRA blends and possible cross reactions between HMN/TETRA blends, as well as further autoignition studies of NC18H38, HMN, and/or their blends at low temperatures. These results will contribute to understanding the autoignition chemistry of diesel surrogates that is vital in reducing the design cycle of future diesel engines.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2020.05.017.

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